خدمتكم طريق خضناه لرضى الله

2025

دفتر المادة

د.كامل فليفل

جبر خطي



لجنة سنافر البوليتكنك

07 9791 3335

Matrix

plural "matrices"

$$A = [A] = \underline{A} = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix}$$

column

row

size of A = dimensions of A = 2×3 → no. of columns

↓ no. of rows

$$B = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 7 & 8 \\ -1 & 0 & 5 \end{bmatrix}$$ 3×3 - matrix

A is $n \times m$ - matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \qquad A = (a_{ij}) \quad 1 \le i \le n, 1 \le j \le m$$

If $n = m$ the A is called square matrix

else, is rectangular matrix.

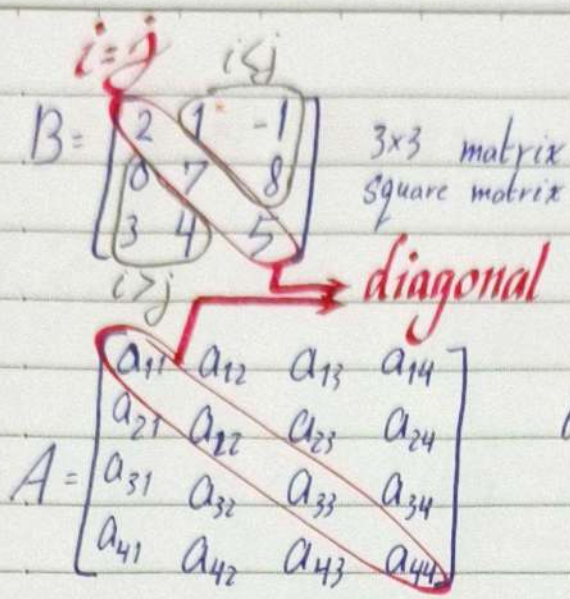
$$A = \begin{bmatrix} 2 & 1 \\ 10 & -1 \\ 3 & 7 \end{bmatrix}$$ 3×2 - matrix

elements (entries)

$a_{11} = 2$, $a_{12} = 1$

$a_{31} = 3$, $a_{32} = 7$

⋮



$$B=\begin{bmatrix}2 & 1 & -1\\ 0 & 7 & 8\\ 3 & 4 & 5\end{bmatrix}$$ 3×3 matrix
square matrix

$$A=\begin{bmatrix}a_{11} & a_{12} & a_{13} & a_{14}\\ a_{21} & a_{22} & a_{23} & a_{24}\\ a_{31} & a_{32} & a_{33} & a_{34}\\ a_{41} & a_{42} & a_{43} & a_{44}\end{bmatrix}$$

$$\text{trace}(A)=\sum_{i=1}^{n} a_{ii}$$

Eg. If $A=[a_{ij}]$ is 3×2 - matrix

if $a_{ij}=\begin{cases} i+j & , i\neq j \\ 0 & , i=j \end{cases}$, write the matrix A

$$A=\begin{bmatrix}0 & 3\\ 3 & 0\\ 4 & 5\end{bmatrix}$$

Eg. If $A=\begin{bmatrix}2 & 1\\ 4 & -3\end{bmatrix}$ find trace of A.

$\text{trace}(A)=2+(-3)=-1$

وَيَوْمَئِذٍ يَفْرَحُ الْمُؤْمِنُونَ بِنَصْرِ اللَّهِ
يَنْصُرُ مَنْ يَشَاءُ وَهُوَ الْعَزِيزُ الرَّحِيمُ

Kinds of Matrix:-

i. Zero matrix

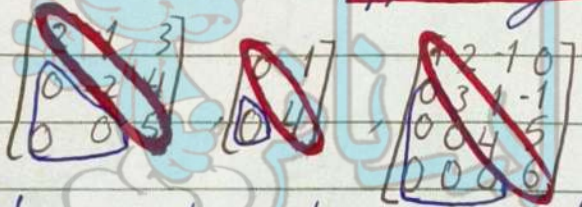
$$0 = [a_{ij}] \text{ where } a_{ij} = 0 \ \forall i,j$$

$$0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

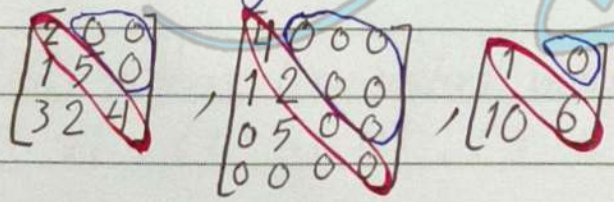
$$0\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad 0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

ii. If $A = [a_{ij}]$, is $n \times n$-matrix (square)

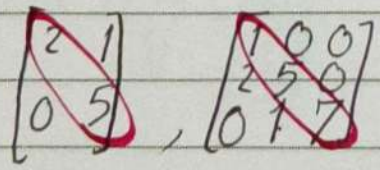
a) A is called a upper triangular matrix, if $a_{ij} = 0 \ \forall i > j$



b) Lower triangular matrix, if $a_{ij} = 0 \ \forall i < j$

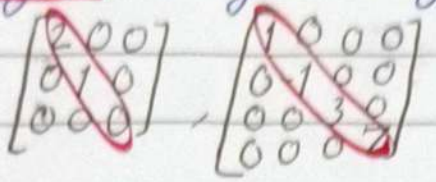


Note: If A is upper or lower triangular matrix then A is called a triangular matrix.

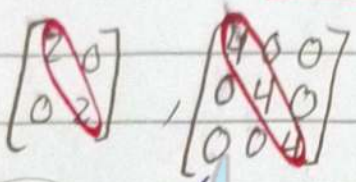


iii. $A=[a_{ij}]$ is $n \cdot n$-matrix then A is

a) diagonal if $a_{ij}=0 \quad \forall i \neq j$



b) is called a scalar matrix if $a_{ij}=\begin{cases} 0, & i \neq j \\ \lambda, & i=j \end{cases}$ λ scalar



iv. Identity matrix / Unit matrix (for multiplication)

it is a scalar matrix with $\lambda=1$

$$I=\begin{bmatrix}1&0\\0&1\end{bmatrix}=I_2, \quad I=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=I_3, \quad I=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}=I_4$$

v. Row matrix (row-vector), matrix with 1 row.

$[1 \;\; 2]$, $[2 \;\; -1 \;\; 4]$, $[-1 \;\; 0 \;\; 1 \;\; 2 \;\; 5]$

vi. Column matrix (column vector), matrix with 1 column.

$$\begin{bmatrix}2\\1\end{bmatrix}, \begin{bmatrix}7\\-1\\3\end{bmatrix}, \begin{bmatrix}-1\\0\\2\\3\end{bmatrix}$$

Def: If $A=[a_{ij}]$ is $n\times m$-matrix then the transpose of A is $A^t=A^T=[a_{ji}]$ is $m\times n$-matrix

$$A=\begin{bmatrix}2&1&-1\\4&5&5\end{bmatrix} \Rightarrow A^T=\begin{bmatrix}2&4\\1&5\\-1&-5\end{bmatrix} \qquad \boxed{(A^T)^T=A}$$

$$B=\begin{bmatrix}5&6\\1&7\end{bmatrix} \Rightarrow B^T=\begin{bmatrix}5&1\\6&7\end{bmatrix}$$

$$C=\begin{bmatrix}2&1&3\\1&0&4\\3&4&7\end{bmatrix} \Rightarrow C^T=\begin{bmatrix}2&1&3\\1&0&4\\3&4&7\end{bmatrix}$$

Def: If A and B are two matricies of the same then $A=B$ if $a_{ij}=b_{ij}\ \forall i,j$

E.g. Given that $\begin{bmatrix}x+1&4&7\\-3&8&5\end{bmatrix}=\begin{bmatrix}6&y^2&7\\-3&2z&5\end{bmatrix}$

Find x,y,z.

$x+1=6 \to x=5$

$y^2=4 \to y=\pm 2$

$2z=8 \to z=4$

Def: A matrix A is called symmetric if $A^T=A$

Eg. Which of the following is a symmetric matrix

1) $A=\begin{bmatrix}2&1&-1\\4&1&0\end{bmatrix}$ $A^T=\begin{bmatrix}2&4\\1&1\\-1&0\end{bmatrix}$ $A^T\neq A$, not symmetric

2) $B=\begin{bmatrix}1&5&4\\5&0&3\\4&3&2\end{bmatrix}$ $B^T=\begin{bmatrix}1&5&4\\5&0&3\\4&3&2\end{bmatrix}$ $B^T=B$, symmetric

3) $A=\begin{bmatrix}1&2\\4&5\end{bmatrix}$ $A^T=\begin{bmatrix}1&4\\2&5\end{bmatrix}$ $A^T\neq A$, not symmetric

Operations on Matrices

Def: If $A=[a_{ij}]$, $B=[b_{ij}]$ are two $n\times m$-matrices then

1) $A+B=[a_{ij}+b_{ij}]$
2) $A-B=[a_{ij}-b_{ij}]$
3) $kA=[ka_{ij}]$ k: scalar

Eg. Find if possible

1) $\begin{bmatrix}2&1&-1\\-2&3&0\end{bmatrix}+\begin{bmatrix}-7&1&8\\0&-3&7\end{bmatrix}=\begin{bmatrix}-5&2&7\\-2&0&7\end{bmatrix}$ addition is comnutitive

2) $\begin{bmatrix}1\\-3\end{bmatrix}+\begin{bmatrix}4\\2\end{bmatrix}=\begin{bmatrix}5\\-1\end{bmatrix}$

3) $\begin{bmatrix}4&1\\-1&7\end{bmatrix}-3\begin{bmatrix}1&0\\2&1\end{bmatrix}=\begin{bmatrix}4&1\\-1&7\end{bmatrix}-\begin{bmatrix}3&0\\6&3\end{bmatrix}=\begin{bmatrix}1&1\\-7&4\end{bmatrix}$

4) $\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ = undefined

5) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}^T + \begin{bmatrix} 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 1 \end{bmatrix}$

Matrix-multiplication

Def: If $A = [a_{ij}]$ is $n \times m$-matrix and
$B = [b_{ij}]$ is $m \times r$-matrix
then $AB = C$
when $c_{ij} = \sum_{s=1}^{m} a_{is} b_{sj}$

Eg. Find if possible

1) $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 5 \\ 2 & -1 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ 10 & 0 \end{bmatrix}$

$2 \times 3 = 3 \times 2$

↳ dimensions of answer matrix

2) $\begin{bmatrix} -1 & 5 \\ 2 & -1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 14 & 6 \\ 4 & -1 & -3 \\ 8 & 13 & -1 \end{bmatrix}$

$3 \times 2 \quad 2 \times 3 \quad 3 \times 3$

$AB \neq BA$

3) $\underset{2\times1}{\begin{bmatrix}2\\1\end{bmatrix}}\underset{2\times1}{\begin{bmatrix}5\\6\end{bmatrix}}$ is undefined

4) $\begin{bmatrix}2\\1\end{bmatrix}\begin{bmatrix}5\\6\end{bmatrix}^T = \underset{2\times1}{\begin{bmatrix}2\\1\end{bmatrix}}\underset{1\times2}{\begin{bmatrix}5 & 6\end{bmatrix}} = \begin{bmatrix}10 & 12\\5 & 6\end{bmatrix}$

5) $\begin{bmatrix}5\\6\end{bmatrix}^T\begin{bmatrix}2\\1\end{bmatrix} = \underset{1\times2}{\begin{bmatrix}5 & 6\end{bmatrix}}\underset{2\times1}{\begin{bmatrix}2\\1\end{bmatrix}} = \begin{bmatrix}16\end{bmatrix}$

6) $\underset{2\times3}{\begin{bmatrix}2 & 1 & 3\\1 & 0 & 2\end{bmatrix}}\underset{3\times3}{\begin{bmatrix}1 & -1 & 5\\0 & 1 & 2\\2 & -1 & 4\end{bmatrix}} = \begin{bmatrix}8 & -4 & 24\\5 & -3 & 13\end{bmatrix}$

7) $\underset{3\times3}{\begin{bmatrix}1 & -1 & 5\\0 & 1 & 2\\2 & -1 & 4\end{bmatrix}}\underset{2\times3}{\begin{bmatrix}2 & 1 & 3\\1 & 0 & 2\end{bmatrix}}$ is undefined

8) $\underset{2\times2}{\begin{bmatrix}a & b\\c & d\end{bmatrix}}\underset{2\times2}{\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}} = \begin{bmatrix}a & b\\c & d\end{bmatrix}$

9) $\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}a & b\\c & d\end{bmatrix} = \begin{bmatrix}a & b\\c & d\end{bmatrix}$

if A is $n\times n$-matrix then $AI_n = I_nA = A$

Eg. Given that $A=\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}$, $B=\begin{bmatrix}2 & 0\\ 3 & -1\end{bmatrix}$

Find:

1) AB $\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}\begin{bmatrix}2 & 0\\ 3 & -1\end{bmatrix}=\begin{bmatrix}-1 & 1\\ 13 & -3\end{bmatrix}$

2) BA $\begin{bmatrix}2 & 0\\ 3 & -1\end{bmatrix}\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}=\begin{bmatrix}2 & -2\\ 1 & -6\end{bmatrix}$

3) A^2 $\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}=\begin{bmatrix}-1 & -4\\ 8 & 7\end{bmatrix}$

4) B^2 $\begin{bmatrix}2 & 0\\ 3 & -1\end{bmatrix}\begin{bmatrix}2 & 0\\ 3 & -1\end{bmatrix}=\begin{bmatrix}4 & 0\\ 3 & 1\end{bmatrix}$

5) $(A+B)^2$ $\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}+\begin{bmatrix}2 & 0\\ 3 & -1\end{bmatrix}=\begin{bmatrix}3 & -1\\ 5 & -2\end{bmatrix}\begin{bmatrix}3 & -1\\ 5 & 2\end{bmatrix}=\begin{bmatrix}4 & -5\\ 25 & 1\end{bmatrix}$

6) $A^2+2AB+B^2$ $\begin{bmatrix}-1 & -4\\ 8 & 7\end{bmatrix}+2\begin{bmatrix}-1 & 1\\ 13 & -3\end{bmatrix}+\begin{bmatrix}4 & 0\\ 3 & 1\end{bmatrix}$

$=\begin{bmatrix}3 & -4\\ 11 & 8\end{bmatrix}+\begin{bmatrix}-2 & 2\\ 26 & -6\end{bmatrix}=\begin{bmatrix}1 & -2\\ 37 & 2\end{bmatrix}$

7) $A^2+AB+BA+B^2$

$\begin{bmatrix}-1 & -4\\ 8 & 7\end{bmatrix}+\begin{bmatrix}-1 & 1\\ 13 & -3\end{bmatrix}+\begin{bmatrix}2 & -2\\ 1 & -6\end{bmatrix}+\begin{bmatrix}4 & 0\\ 3 & 1\end{bmatrix}=\begin{bmatrix}4 & -5\\ 25 & -1\end{bmatrix}$

8) $(AB)^T \begin{bmatrix} -1 & 1 \\ 13 & -3 \end{bmatrix}^T = \begin{bmatrix} -1 & 13 \\ 1 & -3 \end{bmatrix}$

9) $A^T B^T = \begin{bmatrix} 2 & 1 \\ -2 & 6 \end{bmatrix}$

$(AB)^T = B^T A^T$
$(ABC)^T = C^T B^T A^T$

10) $B^T A^T = \begin{bmatrix} -1 & 13 \\ 1 & -3 \end{bmatrix}$

if $C = \begin{bmatrix} -1 & 5 \\ 2 & -1 \end{bmatrix}$, find

11) $(A+B)+C = \begin{bmatrix} 3 & -1 \\ 5 & -2 \end{bmatrix} + \begin{bmatrix} -1 & 5 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 7 & -3 \end{bmatrix}$

12) $A+(B+C) = \begin{bmatrix} 2 & 4 \\ 7 & -3 \end{bmatrix}$

13) $(AB)C \quad \begin{bmatrix} -1 & 1 \\ 13 & -3 \end{bmatrix} \begin{bmatrix} -1 & 5 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ -19 & 68 \end{bmatrix}$

14) $A(BC) \quad \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 5 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 10 \\ -5 & 16 \end{bmatrix}$

$\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -2 & 10 \\ -5 & 16 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ -19 & -68 \end{bmatrix}$

15) $A(B+C)$ $\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}\begin{bmatrix}1 & 5\\ 5 & -2\end{bmatrix} = \begin{bmatrix}-4 & 7\\ 17 & 4\end{bmatrix}$

16) $AB+AC$ $\begin{bmatrix}-4 & 7\\ 17 & 4\end{bmatrix}$

17) $A(B-C)$
18) $AB-AC$
19) $(2A)^T$
20) $2A^T$

Properties

1) $A+0=A=0+A$
2) $A+B=B+A$ (commutative)
3) $A+(B+C)=(A+B)+C$ (associative)
4) $k(A+B)=kA+kB$
5) $(k+l)A=kA+lA$
6) $-1A=-A$
7) $A+(-A)=0$
8) $A(B\mp C)=AB\mp AC$ (distribution)
9) $AI=IA=A$
10) $A(BC)=(AB)C$ (associative)
11) $k(AB)=(kA)B=A(kB)$
12) $(AB)^T=B^TA^T$

$AB\neq BA$

Determinant (for square matrices)

Def: $A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then determinant of A is $\det(A)=|A|=a_{11}a_{22}-a_{12}a_{21}$

Note: If A is $n\times n$-matrix

then if $\det(A)=0$, then A is called → singular, non-invertible, has no inverse

but if $\det(A)\neq 0$ then A is → non-singular, invertible, has inverse

Eg. Find the determinant of

1) $A=\begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$ $\det(A)=(1)(-1)-(2)(3)=-7$

2) $B=\begin{bmatrix} 8 & 2 \\ 4 & 1 \end{bmatrix}$ $\det(A)=8-8=0$ (singular)

3) $C=\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ $\det(C)=\cos^2\theta+\sin^2\theta=1$ (non-singular)

Eg. Find

$\begin{bmatrix} 5 & 1 \\ -1 & 6 \end{bmatrix}$ $30-(-1)=31$

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \quad (-1)^{i+j} \quad \begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

Eg. Find the determinant of $A = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 4 & 0 \\ 5 & -1 & 2 \end{bmatrix}$

method I

$$\det(A) = 2\begin{vmatrix} 4 & 0 \\ -1 & -2 \end{vmatrix} - (1)\begin{vmatrix} -1 & 0 \\ 5 & 2 \end{vmatrix} + 3\begin{vmatrix} -1 & 4 \\ 5 & -1 \end{vmatrix}$$

$$= 2(8-0) - 1(-2-0) + 3(1-20) = -39$$

method II

$$\begin{vmatrix} 2 & 1 & 3 \\ -1 & 4 & 0 \\ 5 & -1 & 2 \end{vmatrix} = +3\begin{vmatrix} -1 & 4 \\ 5 & -1 \end{vmatrix} - 0 + 2\begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} = 3(1-20) + 2(8+1) = -39$$

Eg. Find $\begin{vmatrix} 2 & 1 & -1 & 0 \\ 3 & 1 & -2 & 0 \\ 1 & 1 & -1 & 2 \\ -1 & 2 & 5 & 1 \end{vmatrix}$

$$= -2\begin{vmatrix} 2 & 1 & -1 \\ 3 & 1 & -2 \\ -1 & 2 & 5 \end{vmatrix} + (-1)\begin{vmatrix} 2 & 1 & -1 \\ 3 & 1 & -2 \\ 1 & 1 & -1 \end{vmatrix} = -2\left(-1\begin{vmatrix} 3 & -2 \\ -1 & 5 \end{vmatrix} + 1\begin{vmatrix} 2 & -1 \\ -1 & 5 \end{vmatrix} - 2\begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix}\right)$$

$$+ -1\left(2\begin{vmatrix} 1 & -2 \\ 1 & -1 \end{vmatrix} - 3\begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} + 1\begin{vmatrix} 1 & -1 \\ 1 & -2 \end{vmatrix}\right)$$

$$= -2(-2) - 1(1) = 3$$

The inverse (if any) of 2×2-matrix

If $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\quad \det(A) = ab - bc$

$=0$ → no inverse $\qquad \neq 0$ → $A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Eg. Find the inverse of the given matrix (if it has)

$I=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

1) $A=\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$ $\quad \det(A) = 2-3 = -1 \neq 0$

$A^{-1} = \frac{1}{-1}\begin{bmatrix} 1 & -3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$

check $A \cdot A^{-1} = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

2) $B=\begin{bmatrix} 2 & 8 \\ 1 & 4 \end{bmatrix}$ $\quad \det(B) = 0$, no inverse

3) $C=\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ $\quad \det(C) = \cos^2 x + \sin^2 x = 1$

$C^{-1} = \frac{1}{1}\begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix} = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$

4) $A=\begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$ $\quad \det(A) = 10-3 = 7$

$A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & -3 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} 2/7 & -3/7 \\ -1/7 & 2/7 \end{bmatrix}$

Propreties of Determinates

1) Any matrix that contains zero row (zero column) its determinate

Eg. Find $\begin{bmatrix} 2 & 1 & 0 \\ 3 & -1 & 0 \\ 5 & 6 & 0 \end{bmatrix} = 0$

2) $\det(A^T) = \det(A)$

Eg. If $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 4 \\ 1 & -1 & 0 \end{bmatrix}$, find a) $\det(A)$ b) $\det(A^T)$

a) $\det(A) = -(1)(4+3) - (-1)(8-0) = 7+8 = 15$

b) $A^T \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & -1 \\ -1 & 4 & 0 \end{bmatrix}$, $\det(A^T) = 2(0-(-4)) + 1(4+3) = 8+7 = 15$

3) Any matrix that contains a row (column) that is a multiple of another row (column) then the determinante of this matrix is zero.

Eg. Find

a) $\begin{bmatrix} 2 & 1 & 5 \\ 0 & 7 & 8 \\ -4 & -2 & -10 \end{bmatrix} = 0$

b) $\begin{vmatrix} 1 & -1 & 2 & 5 \\ 7 & 2 & -2 & 35 \\ 0 & 8 & 9 & 0 \\ 4 & 8 & 100 & 20 \end{vmatrix} = 20$

4) If B is obtained of A after interchanging 2 rows (2 columns) then $\det(B) = -\det(A)$

Eg. if $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 4 \\ 1 & -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 3 & 4 \\ 2 & 1 & -1 \\ 1 & -2 & 0 \end{bmatrix}$

Find a) $\det(A)$ b) $\det(B)$

a) $\det(A) = 2(0+8) + 1(4+3) = 16+7 = 23$

b) $\det(B) = -3(0+1) + 4(-4-1) = -3-20 = -23$

5) If B is obtained from A by multiplying a row (column) by the constant K then $\det(B) = K\det(A)$

Eg. If $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 2 \\ 4 & 5 \end{bmatrix}$, find $\det(A)$ and $\det(B)$

$\det(A) = 6$

$\det(B) = 12$

$\det(B) = 2\det(A)$

Eg. Given that $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = 10$

a) $\begin{bmatrix} 2a & 2b & 2c \\ 3d & 3e & 3f \\ g & h & i \end{bmatrix} = 60$

b) $\begin{bmatrix} b & a & c \\ e & d & f \\ h & g & i \end{bmatrix} = -10$

c) $\begin{bmatrix} 2d & 2e & 2f \\ g & h & i \\ a & b & c \end{bmatrix} = -1(-1)(2)\ \det(A)$
$= 20$

Eg. If A is 4×4 matrix $\det(A) = 5$ find $\det(2A)$

$(2^4)(5) = 80$

if A is $n \times n$-matrix
then $\det(kA) = k^n \det(A)$

6) If A is a triangular (diagonal) matrix then $\det(A)$ equals the product of the element on the diagonal

Eg. Find a) $\begin{bmatrix} 2 & 3 & 1 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{bmatrix} = (2)(5)(-1) = -10$

b) $\begin{vmatrix} -1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 4 \end{vmatrix} = (-1)(3)(10)(4) = 120$

c) $\det(I) = 1$

7) $\det(AB) = \det(A)\det(B)$

Eg. If $A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}$, find a) $\det(AB)$ b) $\det(A)\det(B)$

$(-2-3)(15-2) = 78$

8) If A is non-singular then $\det(A^{-1}) = \frac{1}{\det(A)}$

proof $A^{-1}A = I$, $\det(A) \neq 0$

$\det((A^{-1})(A)) = \det(I)$

$\det(A^{-1})\det(A) = 1$

$\det(A^{-1}) = \frac{1}{\det(A)}$

Eg. Given that $\det(A) = 2$, $\det(B) = 5$, find:
(such that A and B are 3×3-matrices)

a) $\det(2A^{-1}B) = 2^3 \det(A^{-1}B)$

$= 2^3 \det(A^{-1})\det(B) = 8\left(\frac{1}{2}\right)5 = 20$

b) $\det(10B^T)$
$= 10^3 \det(B^T)$
$= 1000 \det(B) = 5000$

c) $\det(\det(B)B) = \det(5B) = 5^3 \det(B)$
$= 5^3(5)$
$= 625$

Eg. If the Wronskian of the functions, f, g and h is

$$W(f,g,h) = W = \begin{bmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{bmatrix}$$ if $f(x) = e^x$, $g(x) = e^{2x}$, $h(x) = x$, $k(x) = 1$

Find a) $W(f,g,h)$
b) $W(f,g,k)$

$$W = \begin{bmatrix} e^x & e^{2x} & x \\ e^x & 2e^{2x} & 1 \\ e^x & 4e^{2x} & 0 \end{bmatrix} = e^x e^{2x} \begin{bmatrix} 1 & 1 & x \\ 1 & 2 & 1 \\ 1 & 4 & 0 \end{bmatrix} = e^{3x} \begin{bmatrix} 1 & 1 & x \\ 1 & 2 & 1 \\ 1 & 4 & 0 \end{bmatrix}$$

$$= e^{3x}[1(1-2x) - 4(1-x)] = e^{3x}[1-2x-4+4x] = (2x-3)e^{3x}$$

9) If B is obtained from A after adding a multiple of A row(column) to another row(column) then $\det(B) = \det(A)$

Eg. If the wronskian of the functions, f, g and h is

$$W(f,g,h) = W = \begin{bmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{bmatrix}$$, if $f(x)=e^x$, $g(x)=e^{2x}$, $h(x)=x$, $k(x)=1$

Find a) $W(f,g,h)$
b) $W(f,g,k)$

$$W = \begin{bmatrix} e^x & e^{2x} & x \\ e^x & 2e^{2x} & 1 \\ e^x & 4e^{2x} & 0 \end{bmatrix}$$

$$= e^x e^{2x} \begin{bmatrix} 1 & 1 & x \\ 1 & 2 & 1 \\ 1 & 4 & 0 \end{bmatrix} = e^{3x} \begin{bmatrix} 1 & 1 & x \\ 1 & 2 & 1 \\ 1 & 4 & 0 \end{bmatrix} =$$

$$= e^{3x}[1(1-2x) - 4(1-x)] = e^{3x}[1-2x-4+4x] = (2x-3)e^{3x}$$

9) If B is obtained from A after adding a multiple of A row (column) to another row (column) then $\det(B) = \det(A)$

Eg. If $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \\ 1 & -1 & -2 \end{bmatrix}$ $\rightarrow$ $B = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \\ 3 & 3 & -4 \end{bmatrix}$ $2R_1 + R_3$

Find $\det(A)$ and $\det(B)$

$$\det(A) = -3(-4-1) + 0 - 4(-1-2) = 15 + 12 = 27$$

$$\det(B) = -3(-8+3) + 0 - 4(3-6) = 15 + 12 = 27$$

More Examples:

eg.1- If A is $n \times n$-matrix show that 1) $A+A^T$ is symmetric
2) AA^T is symmetric

$$A=\begin{bmatrix}1 & 5\\ -1 & 4\end{bmatrix}, \quad A+A^T=\begin{bmatrix}1 & 5\\ -1 & 4\end{bmatrix}+\begin{bmatrix}1 & -1\\ 5 & 4\end{bmatrix}=\begin{bmatrix}2 & 4\\ 4 & 8\end{bmatrix}$$

$$(A+A^T)^T=\begin{bmatrix}2 & 4\\ 4 & 8\end{bmatrix} \quad A+A^T \text{ is symmetric}$$

Let A be $n \times n$-matrix

1) $(A+A^T)^T = A^T+(A^T)^T$
$= A^T+A$
$= A+A^T$ $\therefore AA^T$ is symmetric

2) $(AA^T)^T = (A^T)^T A^T$
$= AA^T$ $\therefore AA^T$ is symmetric

eg.2- If A, B, C are three matrices, A is non-singular
if $AB=AC \rightarrow B=C$

$-AB=AC$
$\rightarrow A$ has inverse
$\rightarrow A^{-1}(AB)=A^{-1}(AC)$
$\rightarrow (A^{-1}A)B=(A^{-1}A)C$
$\rightarrow IB=IC$
$\rightarrow B=C$

eg.3- If A, B, C are three matrices, such that ($AB=AC$ then $B=C$) prove or disprove.

$A=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$, $B=\begin{bmatrix}2 & 3\\ 1 & 4\end{bmatrix}$, $C=\begin{bmatrix}1 & 2\\ 2 & 5\end{bmatrix}$ $B \neq C$ but $AB=AC$

$AB=\begin{bmatrix}3 & 7\\ 3 & 7\end{bmatrix}$ $AC=\begin{bmatrix}3 & 7\\ 3 & 7\end{bmatrix}$

eg.4- If $p(x)=x^2+2x+3$ (matrix polynomial), $A=\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}$ find $P(A)$.

$$P(A)=A^2+2A+3I$$

System of linear equations

Def: The following linear equation

$$a_{11}x_1+a_{12}x_2+\dots+a_{1n}x_n=b_1$$
$$a_{21}x_1+a_{22}x_2+\dots+a_{2n}x_n=b_2$$
$$\vdots$$
$$a_{m1}x_1+a_{m2}x_2+\dots+a_{mn}x_n=b_m$$

m-equations with n-unknowns (variables) are called system of linear equation.

if $b_1=b_2=\dots=b_m=0$ then the system is called homogeneous, else it is non-homogeneous.

The above system can be written using matrices in two ways.

1)
$$\begin{bmatrix}a_{11} & a_{12} & \dots & a_{1n}\\ a_{21} & a_{22} & \dots & a_{2n}\\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ \vdots\\ x_n\end{bmatrix}=\begin{bmatrix}b_1\\ b_2\\ \vdots\\ b_m\end{bmatrix}$$

$$AX=B$$

$m\times n$ $\quad n\times 1$ $\quad m\times 1$

matrix of coefficient (A) — matrix of unknowns — matrix of constants

2) Augmented matrix $[A:B]$

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1m} & b_1 \\ a_{21} & a_{22} & & a_{2m} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} & b_m \end{array}\right]$$

Eg. Write the given system of linear equations using matrices in two ways.

$$x+y-z=3$$
$$3x+z=8$$

1) $\underbrace{\begin{bmatrix} 1 & 1 & -1 \\ 3 & 0 & 1 \end{bmatrix}}_{A_{2\times 3}} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{X_{3\times 1}} = \underbrace{\begin{bmatrix} 3 \\ 8 \end{bmatrix}}_{B_{2\times 1}}$

2) $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 3 \\ 3 & 0 & 1 & 8 \end{array}\right]$

If $AX=B$ is a system of linear equations

$[A:B]$

The system has unique solution

$x+y=1$

$x-y=5$

$x=3, y=-2$

infinite solution

$x+y=1$

$2x+2y=2$

$(1,0), (0,1)$

$(2,-1), \ldots$

No solution

$x+y=1$

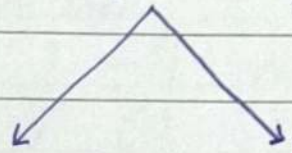
$x+y=7$

The system is consistent

The system is consistent

If $AX = 0$ homogeneous system

$[A : 0]$



unique

$x_1 = 0, x_2 = 0, \ldots$

Trivial solution

infinite solution

non-trivial solution

Def: A matrix is called in the reduced row echelon form (RREF), if it satisfies:

1) zero row(s) (if any) at the bottom.
2) The first element in every non-zero row must be 1 (called leading 1)
3) For any two rows the leading 1 of the lower row must be to the right of the leading 1 of the above.
4) All the elements of the column that contains a leading 1 must be zeros (except the leading)

But if a matrix satisfies only 1, 2 and 3 it is called row echelon form (REF)

Eg. Which of the following matrices in RREF, REF or neither.

1) $\begin{bmatrix} 0 & 1 & 0 & 3 & -1 \\ 0 & 0 & 1 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ in RREF and REF

2) $\begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ in REF, but not in RREF

3) $\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ in RREF

4) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ not in REF, neither not in RREF

5) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ in RREF

وَقَالَ رَبُّكُمُ ادْعُونِي أَسْتَجِبْ لَكُمْ ۚ إِنَّ الَّذِينَ
يَسْتَكْبِرُونَ عَنْ عِبَادَتِي سَيَدْخُلُونَ جَهَنَّمَ دَاخِرِينَ

Elementary Row Operations

i. Interchanging two rows.
ii. Multiplying a row by non-zero constant
iii. Multiplying a row by a constant and adding the result to another row.

Gauss-Jordan Method

To solve any system of linear equation

1) $[A:B]$
   ↓ elementary row operations
2) RREF
3) We get the result

Def: If B is obtained from A using elementary row operation(s), then B is called row equivalent to A.

Eg. Using Gauss-Jordan solve

$2x-y+z=3,\ x+y+z=6,\ 4x+y=6,\ 5x+2y+z=12$

$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 3 \\ 1 & 1 & 1 & 6 \\ 4 & 1 & 0 & 6 \\ 5 & 2 & 1 & 12 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 4 & 1 & 0 & 6 \\ 5 & 2 & 1 & 12 \end{array}\right] \begin{matrix} R_2 \\ R_1 \\ \\ \end{matrix} \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & -3 & -4 & -18 \\ 0 & -3 & -4 & -18 \end{array}\right] \begin{matrix} \\ -2R_1+R_2 \\ -4R_1+R_3 \\ -5R_1+R_4 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & \frac{1}{3} & 3 \\ 0 & -3 & -4 & -18 \\ 0 & -3 & -4 & -18 \end{bmatrix} \begin{matrix} \\ -\frac{1}{3}R_2 \\ \\ \end{matrix} \sim \begin{bmatrix} 1 & 0 & \frac{2}{3} & 3 \\ 0 & 1 & \frac{1}{3} & 3 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & -3 & -9 \end{bmatrix} \begin{matrix} -R_2+R_1 \\ \\ 3R_2+R_3 \\ 3R_2+R_4 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & \frac{2}{3} & 3 \\ 0 & 1 & \frac{1}{3} & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & -3 & -9 \end{bmatrix} \begin{matrix} \\ \\ \\ -\frac{1}{3}R_3 \end{matrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} -\frac{2}{3}R_3+R_1 \\ -\frac{1}{3}R_3+R_2 \\ \\ 3R_3+R_4 \end{matrix}$$

RREF

$x=1$, $y=2$, $z=3$

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Eg. Solve by Gauss-Jordan $x+y+z=6$

$2x+y-z=4$

$3x-y+z=6$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 2 & 1 & -1 & 4 \\ 3 & -1 & 1 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -8 \\ 0 & -4 & -2 & -12 \end{bmatrix} \begin{matrix} \\ -2R_1+R_2 \\ -5R_1+R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 3 & 8 \\ 0 & -4 & -2 & -12 \end{bmatrix} \begin{matrix} \\ -R_2 \\ \\ \end{matrix} \sim \begin{bmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & 10 & 20 \end{bmatrix} \begin{matrix} -R_2+R_1 \\ \\ 4R_2+R_3 \end{matrix}$$

$$\sim\begin{bmatrix}1&0&-2&-2\\0&1&3&8\\0&0&1&2\end{bmatrix}\frac{1}{10}R_3 \qquad \sim\begin{bmatrix}1&0&0&2\\0&1&0&2\\0&0&1&2\end{bmatrix}\begin{matrix}2R_3+R_1\\-3R_3+R_2\end{matrix}$$

$x=2, y=2, z=2$

$$x=\begin{bmatrix}2\\2\\2\end{bmatrix}$$ unique solution

Eg. By Gauss-Jordan, solve

$$x_1+x_2+3x_3+x_4=6$$
$$2x_1-x_2+x_4=2$$
$$x_1+2x_3+x_4=4$$
$$2x_1+x_2+5x_3+2x_4=10$$

$$\begin{bmatrix}1&1&3&1&6\\2&0&-1&1&2\\1&0&2&1&4\\2&1&5&2&10\end{bmatrix}\sim\begin{bmatrix}1&1&3&1&6\\0&-2&-7&-1&-10\\0&-1&-1&0&-2\\0&-1&-1&0&-2\end{bmatrix}\begin{matrix}\\-2R_1+R_2\\-R_1+R_3\\-2R_1+R_4\end{matrix}$$

$$\begin{bmatrix}1&1&3&1&6\\0&1&1&0&2\\0&-2&-7&-1&-10\\0&-1&-1&0&-2\end{bmatrix}\begin{matrix}\\-R_2\\R_2\\\end{matrix}\sim\begin{bmatrix}1&0&2&1&4\\0&1&1&0&2\\0&0&-5&-1&-6\\0&0&0&0&0\end{bmatrix}\begin{matrix}-R_2+R_1\\\\2R_2+R_3\\R_2+R_4\end{matrix}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 4 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & \frac{1}{5} & \frac{6}{5} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ \\ \\ -\frac{1}{5}R_3 \end{array} \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{3}{5} & 8/5 \\ 0 & 1 & 0 & -\frac{1}{5} & 4/5 \\ 0 & 0 & 1 & 1/5 & 6/5 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} -2R_3+R_1 \\ -R_3+R_2 \\ \\ \end{array}$$

$x_1 + \frac{3}{5}x_4 = \frac{8}{5}$, let $x_4 = r \in \mathbb{R} \Rightarrow$

$x_2 - \frac{1}{5}x_4 = 4/5$, $x_1 + \frac{3}{5}r = \frac{8}{5} \Rightarrow x_1 = \frac{8}{5} - \frac{3}{5}r$

$x_3 + \frac{1}{5}x_4 = \frac{6}{5}$, $x_2 = \frac{4}{5} + \frac{1}{5}r$

$x_3 = \frac{6}{5} - \frac{1}{5}r$

$x_4 = r$

$r = -4 \rightarrow x_1 = 4$

$\rightarrow x_2 = 0$

$\rightarrow x_3 = 2$

$\rightarrow x_4 = -4$

$$x = \begin{bmatrix} \frac{8}{5} - \frac{3}{5}r \\ \frac{4}{5} + \frac{1}{5}r \\ \frac{6}{5} - \frac{1}{5}r \\ r \end{bmatrix} = \begin{bmatrix} 8/5 \\ 4/5 \\ 6/5 \\ 0 \end{bmatrix} + r \begin{bmatrix} -\frac{3}{5} \\ +\frac{1}{5} \\ -\frac{1}{5} \\ 1 \end{bmatrix}, r \in \mathbb{R}$$

Eg. By Gauss-Jordan, solve

$x_1 + x_2 + x_3 + x_4 = 4$

$x_1 + 2x_2 + x_4 = 4$

$2x_1 + 3x_2 + x_3 + 2x_4 = 10$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 4 \\ 2 & 3 & 1 & 2 & 10 \end{array}\right] \sim \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 2 \end{array}\right] \begin{array}{l} \\ -R_1+R_2 \\ -2R_1+R_3 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 4 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right] \begin{array}{l} -R_2+R_1 \\ \\ -R_2+R_3 \end{array}$$

$0x_1 + 0x_2 + 0x_3 + 0x_4 = 2$

$0 = 2$ contradiction

$\therefore$ The system has no solution

Eg. Solve $x_1+x_2+x_4+x_5=4$
$x_1+2x_2+x_5=4$

$$\left[\begin{array}{ccccc|c} 1&1&0&1&1&4\\ 1&2&0&0&1&4 \end{array}\right] \sim \left[\begin{array}{ccccc|c} 1&1&0&1&1&4\\ 0&1&0&-1&0&0 \end{array}\right] -R_1+R_2$$

$$\sim \left[\begin{array}{ccccc|c} 1&0&0&2&1&4\\ 0&1&0&-1&0&0 \end{array}\right] -R_2+R_1$$

let $x_4=r$, $x_5=s$, $x_3=t$, $r,s,t\in R$

$x_1+2x_4+x_5=4$ | $x_2=r$

$x_2-x_4=0$ | $x_1=4-2r-s$

$$x=\begin{bmatrix} 4-2r-s\\ r\\ t\\ r\\ s \end{bmatrix} = \begin{bmatrix} 4\\0\\0\\0\\0 \end{bmatrix} + \begin{bmatrix} -2\\1\\0\\1\\0 \end{bmatrix} r + \begin{bmatrix} -1\\0\\0\\0\\1 \end{bmatrix} s + \begin{bmatrix} 0\\0\\1\\0\\0 \end{bmatrix} t$$

1) $A^{-1} = \frac{1}{det(A)} adj(A)$

$A^{-1} det(A) = adj(A)$

$A A^{-1} det(A) = A\, adj(A)$

$A\, adj(A) = det(A)\, I$

2) $A\, adj(A) = \begin{bmatrix} 9 & 0 & 0 \\ x & z & 0 \\ 0 & 0 & 2w \end{bmatrix}$

$x = 0$

$det(A) = 9$

$z = 9$

$2w = 9 \Rightarrow w = \frac{9}{2}$

نهاية مادة البث

General Vector Space

Def: If V is a non-empty set of objects. then V with two operations $\oplus, \odot$ is called a vector space if for any $\vec{u}, \vec{v}$ and $\vec{w} \in \vec{V}$ and for any $k, m \in \mathbb{R}$ satisfies the following axioms:

1. $\vec{u} + \vec{v} \in \vec{V}$ (closure property)
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ (commutative property)
3. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} (\vec{v} + \vec{w})$ (associative)
4. $\exists \vec{0} \in V$ such that $\vec{u} + \vec{0} = \vec{u}$
5. $\forall \vec{u} \in V$, $\exists -\vec{u} \in V$ such that $\vec{u} + (-\vec{u}) = \vec{0}$

6. $k\vec{u} \in V$ (closed under scalar multiplication)
7. $k(\vec{u}+\vec{v}) = k\vec{u}+k\vec{v}$
8. $(k+m)\vec{u} = k\vec{u}+m\vec{u}$
9. $k(m\vec{u}) = (km)\vec{u}$
10. $1\vec{u} = \vec{u}$

Eg. If $V = R^2$
$= \{(x,y) / x, y \in R\}$
$\vec{u} = (u_1, u_2)$; $\vec{v} = (v_1, v_2) \in V$, such that $\vec{u}+\vec{v} = (u_1+v_1, u_2+v_2)$ and $\forall k \in R$, $k\vec{u} = (ku_1, ku_2)$. Show that V under the two operations is a vector space.

1) Let $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2) \in R^2 = V$
$\vec{u}+\vec{v} = (u_1+v_1, u_2+v_2) \in V = R^2$

2) $\vec{u}+\vec{v} = (u_1+v_1, u_2+v_2)$
$= (v_1+u_1, v_2+u_2)$
$= (v_1, v_2) + (u_1+u_2)$
$= \vec{v}+\vec{u}$

3) $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$, $\vec{w} = (w_1, w_2)$
$(\vec{u}+\vec{v})+\vec{w} = (u_1+v_1, u_2+v_2) + (w_1, w_2)$
$= ((u_1+v_1)+w_1, (u_2+v_2)+w_2)$
$= (u_1+(v_1+w_1), u_2+(v_2+w_2))$
$= (u_1+u_2) + (v_1+w_1, v_2+w_2)$
$= \vec{u} + ((v_1, v_2) + (w_1+w_2))$
$= \vec{u} + (\vec{v}+\vec{w})$

4) $\exists \vec{0} = (0,0) \in V = R^2$

$$\vec{u} + \vec{0} = (u_1, u_2) + (0,0)$$
$$= (u_1 + 0, u_2 + 0)$$
$$= (u_1, u_2)$$
$$= \vec{u}$$

5) $\vec{u} = (u_1, u_2)$, $\exists -\vec{u} = (-u_1, -u_2)$

$$= (u_1 + (-u_1), u_2 + (-u_2))$$
$$= (0,0) = \vec{0}$$

6) $k\vec{u} = k(u_1, u_2)$

$$= (ku_1, ku_2) \in V = R^2$$

7) $k(\vec{u} + \vec{v}) = k\left((u_1, u_2) + (v_1, v_2)\right)$

$$= k(u_1 + v_1, u_2 + v_2)$$
$$= \left(k(u_1 + v_1), k(u_2 + v_2)\right)$$
$$= (ku_1 + kv_1, ku_2 + kv_2)$$
$$= (ku_1, ku_2) + (kv_1, kv_2)$$
$$= k(u_1, u_2) + k(v_1, v_2)$$
$$= k\vec{u} + k\vec{v}$$

8) $(k+m)\vec{u} = (k+m)(u_1, u_2)$

$$= \left((k+m)u_1, (k+m)u_2\right)$$
$$= (ku_1 + mu_1, ku_2 + mu_2)$$
$$= (ku_1, ku_2) + (mu_1 + mu_2)$$
$$= k(u_1, u_2) + m(u_1 + u_2)$$
$$= k\vec{u} + m\vec{u}$$

9) $k(m\vec{u}) = k(m(u_1, u_2))$
$= k(mu_1, mu_2)$
$= (k(mu_1), k(mu_2))$
$= ((km)u_1, (km)u_2)$
$= (km)(u_1, u_2)$
$= (km)\vec{u}$

10) $1\vec{u} = 1(u_1, u_2)$
$= (1u_1, 1u_2)$
$= (u_1, u_2)$
$= \vec{u}$

$\therefore V = R^2$ is a vector space under standard two operation

- $V = R^n$ under the usual operation is a vector space
Also $V = R^\infty = \{(x_1, x_2, x_3, \ldots) \; x_1, x_2, \ldots \in R\}$

Eg. If $V = M_{22}$
$= \left\{\begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}, u_1, u_2, u_3, u_4 \in R\right\}$

Let $\vec{u} = \begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$

$\vec{u} + \vec{v} = \begin{bmatrix} u_1+v_1 & u_2+v_2 \\ u_3+v_3 & u_4+v_4 \end{bmatrix}$ // $k\vec{u} = \begin{bmatrix} ku_1 & ku_2 \\ ku_3 & ku_4 \end{bmatrix}$

Let $\vec{u} = \begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}$, $\vec{v} \begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$ and $\vec{w} = \begin{bmatrix} w_1 & w_2 \\ w_3 & w_4 \end{bmatrix} \in V$

$k, m \in R$

1) $\vec{u}+\vec{v}=\begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}+\begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$

$=\begin{bmatrix} u_1+v_1 & u_2+v_2 \\ u_3+v_3 & u_4+v_4 \end{bmatrix} \in V$

2) $\vec{u}+\vec{v}=\begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}+\begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$

$=\begin{bmatrix} u_1+v_1 & u_2+v_2 \\ u_3+v_3 & u_4+v_4 \end{bmatrix}$

$=\begin{bmatrix} v_1+u_1 & v_2+u_2 \\ v_3+u_3 & v_4+u_4 \end{bmatrix}$

$=\begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}+\begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}$

$=\vec{v}+\vec{u}$

3) $(\vec{u}+\vec{v})+\vec{w}=\vec{u}+(\vec{v}+\vec{w})$ check

4) $\vec{0}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\vec{u}+\vec{0}=\vec{u}$

5) $\vec{u}=\begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix} \quad -\vec{u}\begin{bmatrix} -u_1 & -u_2 \\ -u_3 & -u_4 \end{bmatrix}$

$\vec{u}+(-\vec{u})=\vec{0}$



Eg. Given that $V = \mathbb{R}^+$

Let $x, y \in \mathbb{R}^+$

$x + y = xy$

Let $k \in \mathbb{R}$, $kx = x^k$

Let $x, y, z \in V = \mathbb{R}^+$

$k, m \in \mathbb{R}$

1) $x + y = xy \in V = \mathbb{R}^+$

2) $x + y = xy$

$= yx = y + x$

3) $(x+y) + z = (xy) + z$

$= (xy)z$

$= x(yz)$

$= x + (y+z)$

4) $\exists \vec{0} = 1 \in V = \mathbb{R}^+$

$x + \vec{0} = x + 1$

$= x1 = x$

5) $\exists -x = \frac{1}{x} \in V$

$x + (-x) = x(-x)$

$= x\frac{1}{x} = 1$

6) $kx = x^k \in \mathbb{R}^+ = V$

7) $k(x+y) = (x+y)^k$

$= (xy)^k$

$= x^k y^k$

$= y^k x^k$

$= y^k + x^k$

$= ky + kx$

8) $(k+m)x = x^{k+m}$

$= x^k x^m$

$= kx \; mx$

$= kx + mx$

9) $k(mx) = (mx)^k$

$= (x^m)^k$

$= x^{mk}$

$= (mk)x$

10) $1x = x^1$

$= x$ $\therefore V = \mathbb{R}^+$ under the given operations is a vector space

Eg. Let $V = F(-\infty, \infty) = \{f \mid f : \mathbb{R} \to \mathbb{R}\}$

$(f+g)(x) = f(x) + g(x)$

$(kf)(x) = kf(x)$

Let $f, g, h \in V = F(-\infty, \infty)$

$k, m \in \mathbb{R}$

1) $D_{f+g} = D_f \cap D_g = \mathbb{R}$ $\therefore f+g \in V$

2) $f+g \stackrel{?}{=} g+f$

$(f+g)(x) = f(x) + g(x)$

$= g(x) + f(x)$

$= (g+f)(x)$ $\therefore f+g = g+f$

3) $(f+g)+h \stackrel{?}{=} f+(g+h)$

$((f+g)+h)(x) = (f+g)(x) + h(x)$

$= (f(x)+g(x)) + h(x)$

$= f(x) + (g(x) + h(x))$

$= f(x) + (g+h)(x)$

$= (f + (g+h))(x)$ $\therefore (f+g)+h = f+(g+h)$

4) $\exists \vec{0} \in F(-\infty, \infty)$

$\vec{0}(x) = 0$

$(f+\vec{0})(x) = f(x) + \vec{0}(x)$

$= f(x) + 0$

$= f(x)$ $\quad \therefore f + \vec{0} = f$

5) $(-f)(x) = -f(x)$

$(f + (-f))(x) = f(x) + (-f)x$

$= f(x) + (-f(x))$

$= 0$

$(f + (-f))(x) = \vec{0}(x)$ $\quad \therefore f + (-f) = \vec{0}$

6) $(kf)(x) = kf(x)$

$kf \in F(-\infty, \infty) = V$

7) $k(f+g)(x) = k(f(x) + g(x))$

$= kf(x) + kg(x)$

$= (kf + kg)(x)$

$k(f+g) = kf + kg$

8) $(k+m)f \stackrel{?}{=} kf + mf$

$((k+m)f)(x) = (k+m)f(x)$

$= kf(x) + mf(x)$

$= (kf + mf)(x)$ $\quad \therefore (k+m)f = kf + mf$

9) $k(mf) \stackrel{?}{=} (km)f$

$(k(mf))(x) = k(mf(x))$

$= (km)f(x)$

$= ((km)f)(x)$ $\quad \therefore k(mf) = (km)f$

10) $1f \stackrel{?}{=} f$

$(1f)(x) = 1\,f(x)$

$= f(x)$ $\quad \therefore 1f = f$

$\therefore F(-\infty, \infty)$ is a vector space

Eg. $V = \left\{ \begin{bmatrix} a & b \\ c & 1 \end{bmatrix} \quad a, b, c \in \mathbb{R} \right\}$

$+ : A + B$ adding matrices

kA: scalar multiplication in matrices

Is V a vector space?

Let $A = \begin{bmatrix} a_1 & b_1 \\ c_1 & 1 \end{bmatrix}$

$B = \begin{bmatrix} a_2 & b_2 \\ c_2 & 1 \end{bmatrix} \in V$

$A + B = \begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & 2 \end{bmatrix} \notin V$

$\therefore V$ is not a vector space

Eg. Let $V = \mathbb{R}^2$

$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$

$k(x, y) = (kx, 0)$

is V a vector space under these operations

$(2, 3) \in V$

$1(2, 3) = (2, 0)$

$\neq (2, 3)$

$\therefore V$ is not a vector space

EX: $V=\{\vec{0}\}$ → Zero vector space أو: null space

$\vec{0}+\vec{0}=\vec{0}$

$k\vec{0}=\vec{0}$

Show that V under these operations is a vector space.

Thm: If V is a vector space $\vec{u}\in V$, $k\in\mathbb{R}$

then 1) $0\vec{u}=\vec{0}$

2) $k\vec{0}=\vec{0}$

3) $(-1)\vec{u}=-\vec{u}$

4) $k\vec{u}=\vec{0} \rightarrow k=0$ or $\vec{u}=\vec{0}$

1) $0=0+0$

$0\vec{u}=(0+0)\vec{u}$

$0\vec{u}=0\vec{u}+0\vec{u}$

$0\vec{u}\in V \rightarrow \exists -0\vec{u}\in V$ such that $0\vec{u}+(-0\vec{u})=\vec{0}$

$0\vec{u}+(-0\vec{u})=(0\vec{u}+0\vec{u})+(-0\vec{u})$

$\vec{0}=0\vec{u}+(0\vec{u}+(-0\vec{u}))$

$\vec{0}=0\vec{u}+\vec{0}$ $\therefore 0\vec{u}=\vec{0}$

2) $\vec{0}+\vec{0}=\vec{0}$

$k(\vec{0}+\vec{0})=k\vec{0}$

$k\vec{0}+k\vec{0}=k\vec{0}$

$-k\vec{0}+(k\vec{0}+k\vec{0})=(-k\vec{0})+k\vec{0}$

$(-k\vec{0}+k\vec{0})+k\vec{0}=\vec{0}$

$\vec{0}+k\vec{0}=\vec{0}$

$k\vec{0}=\vec{0}$

3) $(1+(-1))=0$

$(1+(-1))\vec{u}=0\vec{u}$

$1\vec{u}+(-1)\vec{u}=\vec{0}$ from ①

$\vec{u}+(-1)\vec{u}=\vec{0}$

$-\vec{u}+(\vec{u}+(-1)\vec{u})=-\vec{u}+\vec{0}$

$(-\vec{u}+\vec{u})+(-1)\vec{u}=-\vec{u}$

$\vec{0}+(-1)\vec{u}=-\vec{u}$

$(-1)\vec{u}=-\vec{u}$

4) $k\vec{u}=\vec{0}$

if $k=0$ done

if $k\neq 0 \rightarrow \frac{1}{k}(k\vec{u})=\frac{1}{k}\vec{0}$

$(\frac{1}{k}k)\vec{u}=\vec{0}$

$1\vec{u}=\vec{0}$

$\vec{u}=\vec{0}$

Vector Subspace

Def: If V is a vector space, W is a non-empty subset of V ($\phi \neq W \leq V$) such that W under the same two operations on V is a vector space, then W is called a subspace of V

Thm: If V is a vector space, W is a non-empty subset of V, then W is a vector subspace of V iff

1) $\forall \vec{u},\vec{v} \in W \rightarrow \vec{u}+\vec{v} \in W$

2) $\forall k \in \mathbb{R}\ \forall \vec{u} \in W \rightarrow k\vec{u} \in W$

Eg. If $V=\mathbb{R}^3$ under the usual operations which of the following is a vector subspace of V

1) $W_1=\{(x,y,0) \mid x,y\in\mathbb{R}\}$

$W_1\neq 0$

let $\vec{u}=(x_1,y_1,0)$, $\vec{v}=(x_2,y_2,0)\in W_1$

$\vec{u}+\vec{v}=(x_1,y_1,0)+(x_2,y_2,0)$

$=(x_1+x_2,y_1+y_2,0)\in W_1$ ✓

let $\vec{u}=(x,y,0)\in W_1$

let $k\in\mathbb{R}$

$k\vec{u}=k(x,y,0)=(kx,ky,0)\in W_1$ ✓

$\therefore W_1$ is a vector subspace of V

2) $W_2=\{(x,y,1) \mid x,y\in\mathbb{R}\}$

let $\vec{u}=(2,1,1)$ $\vec{v}=(3,4,1)$

$\vec{u}+\vec{v}=(5,5,2)\notin W_2$

$\therefore W_2$ is not a vector subspace

3) $W_3=\{(a,b,a+b) \mid a,b\in\mathbb{R}\}$

let $\vec{x}=(a,b,a+b)$, $\vec{y}=(c,d,c+d)$

$\vec{x}+\vec{y}=(a+c,b+d,(a+b)+(c+d))$

$=(a+b,b+d,(a+c)+(b+d))$

$\therefore \vec{x}+\vec{y}\in W_3$

let $k\in\mathbb{R}$, $\vec{x}=(a,b,a+b)\in W_3$

$k\vec{x}=k(a,b,a+b)$

$=(ka,kb,k(a+b))=(ka,kb,ka+kb)\in W_3$

$\therefore W_3$ is a vector subspace of V

Subspaces

Thm: Let V be a vector space, $\vec{u} \in V$, $k \in \mathbb{R}$, then

a) $0\vec{u} = \vec{0}$

b) $k\vec{0} = \vec{0}$

c) $(-1)\vec{u} = -\vec{u}$

d) If $k\vec{u} = \vec{0}$ then $k = 0$ or $\vec{u} = \vec{0}$.

Proof:

a) $0 = 0 + 0$

$\Rightarrow 0\vec{u} = (0+0)\vec{u}$

$\Rightarrow 0\vec{u} = 0\vec{u} + 0\vec{u}$ axiom 8

since $0\vec{u} \in V$ (axiom 6)

$\Rightarrow \exists\ -0\vec{u} \in V$ such that $0\vec{u} + (-0\vec{u}) = \vec{0}$

$\therefore -0\vec{u} + 0\vec{u} = -0\vec{u} + (0\vec{u} + 0\vec{u})$

$\vec{0} = (-0\vec{u} + 0\vec{u}) + 0\vec{u}$ axiom 3

$\vec{0} = \vec{0} + 0\vec{u}$

$\vec{0} = 0\vec{u}$ axiom 4

$0\vec{u} = \vec{0}$.

b) $k\vec{0} = \vec{0}$

$\vec{0} + \vec{0} = \vec{0}$ ~~axiom 4~~ axiom 4

$\Rightarrow k(\vec{0} + \vec{0}) = k\vec{0}$

$\Rightarrow k\vec{0} + k\vec{0} = k\vec{0}$ axiom 7

$\Rightarrow$ since $k\vec{0} \in V \Rightarrow \exists -k\vec{0} \in V$ such that $k\vec{0} + (-k\vec{0}) = \vec{0}$

$\therefore (k\vec{0} + k\vec{0}) + (-k\vec{0}) = (-k\vec{0}) + k\vec{0}$

$k\vec{0} + (k\vec{0} + (-k\vec{0})) = \vec{0}$ axiom 3

$k\vec{0} + \vec{0} = \vec{0}$

$k\vec{0} = \vec{0}$ axiom 4

c) $(-1)\vec{u} = -\vec{u}$

$-1 + 1 = 0$

$\Rightarrow (-1+1)\vec{u} = 0\vec{u}$

$(-1)\vec{u} + 1\vec{u} = \vec{0}$ by (a) and axiom 8

$(-1)\vec{u} + \vec{u} = \vec{0}$ axiom 10

since $\vec{u} \in V \Rightarrow \exists -\vec{u} \in V$ such $\vec{u} + (-\vec{u}) = \vec{0}$ axiom 5

$\therefore ((-1)\vec{u} + \vec{u}) + (-\vec{u}) = \vec{0} + (-\vec{u})$

$(-1)\vec{u} + (\vec{u} + (-\vec{u})) = -\vec{u}$ axiom 3, axiom 4

$(-1)\vec{u} + \vec{0} = -\vec{u}$

$\therefore (-1)\vec{u} = -\vec{u}$

d) If $k\vec{u} = \vec{0}$ then $k = 0$ or $\vec{u} = \vec{0}$

if $k = 0$ done, but if $k \neq 0$

$\Rightarrow \frac{1}{k}(k\vec{u}) = \frac{1}{k}\vec{0}$

$\rightarrow (\frac{1}{k}k)\vec{u} = \vec{0}$ from (b) and axiom 9

$\rightarrow 1\,\vec{u} = \vec{0}$

$\rightarrow \vec{u} = \vec{0}$ axiom 10

Subspaces:

Def: If V is a vector space, $\phi \neq W \subseteq V$ then W is called a vector subspace of V, if W itself is a vector space under the addition and scalar multiplication defined on V.

Thm: If W is a non-empty subset of V then W is a subspace of V iff

a) $\forall \vec{u}, \vec{v} \in W \Rightarrow \vec{u} + \vec{v} \in W$

b) $\forall k \in \mathbb{R} \ \forall \vec{u} \in W \Rightarrow k\vec{u} \in W$.

E.g: If $V = \mathbb{R}^3$ is the Eucledian vector space (under the standard operations)

$$W = \{ \vec{u} = (u_1, u_2, 0) \mid u_1, u_2 \in \mathbb{R} \},$$

Show that W is a subspace of V

Sol:

Let $\vec{u} = (u_1, u_2, 0)$, $\vec{v} = (v_1, v_2, 0) \in W$

$\Rightarrow \vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, 0) \in W$

Let $k \in \mathbb{R}$, $\vec{u} = (u_1, u_2, 0) \in V \rightarrow k\vec{u} = (ku_1, ku_2, 0) \in W$

$\therefore W$ is a subspace of V

e.g: If V is vector space then

1) $W = \{\vec{0}\}$ is subspace of V

2) V is a subspace of V.

Sol:

1) $W = \{\vec{0}\}$

Let $\vec{0} \in W \Rightarrow \vec{0} + \vec{0} = \vec{0} \in W$

Let $k \in R, \vec{0} \in W \Rightarrow k\vec{0} = \vec{0} \in W$

$\therefore W$ is a subspace of V.

2) trivial.

e.g: If $C(-\infty, \infty) = \{f \mid f \text{ is continuous on } R\}$

then $C(-\infty, \infty)$ is a subspace of $F(-\infty, \infty)$

Proof

Let $f, g \in C(-\infty, \infty) \Rightarrow f+g$ is continuous on R

$\Rightarrow f+g \in C(-\infty, \infty)$

Let $k \in R, f \in C(-\infty, \infty) \Rightarrow kf$ is continuous on R

$\Rightarrow kf \in C(-\infty, \infty)$

$\therefore C(-\infty, \infty)$ is a subspace of $F(-\infty, \infty)$

e.g:

Let $P_n = \{ \vec{P} \mid \vec{P} = a_0 + a_1 x + \cdots + a_n x^n \mid a_0, a_1, \ldots, a_n \in \mathbb{R} \}$

= the set of all polynomials of degree $\leq n$.

then P_n is a subspace of $F(-\infty, \infty)$

Proof:

Let $\vec{P}, \vec{Q} \in P_n$ such that

$\vec{P}(x) = a_0 + a_1 x + \cdots + a_n x^n$,

$Q(x) = b_0 + b_1 x + \cdots + b_n x^n$

then $(\vec{P} + \vec{Q})(x) = \vec{P}(x) + Q(x)$

$= (a_0 + a_1 x + \cdots + a_n x^n) + (b_0 + b_1 x + \cdots + b_n x^n)$

$= (a_0 + b_0) + (a_1 + b_1) x + \cdots + (a_n + b_n) x^n \in P_n$

Now Let $k \in \mathbb{R}$, $\vec{f} = a_0 + a_1 x + \cdots + a_n x^n$

$\Rightarrow k\vec{f} = k(a_0 + a_1 x + \cdots + a_n x^n)$

$= (ka_0) + (ka_1) x + \cdots + (ka_n) x^n \in P_n$

$\therefore P_n$ is subspace of $F(-\infty, \infty)$

Notations:

* $C^1(-\infty,\infty)$ = the set of all functions which their first derivatives are continuous on $\mathbb{R}$

* $C^2(-\infty,\infty)$ = the set of all functions which their second derivatives are continuous on $\mathbb{R}$.

* $C^m(-\infty,\infty)$ = the set of all fuctions which their m^{th} derivatives are continuous on $\mathbb{R}$.

e.g: For any $m=1,2,\dots$, $C^m(-\infty,\infty)$ is a subspace of $F(-\infty,\infty)$

Sol:

Let $\vec{f},\vec{g}\in C^m(-\infty,\infty) \longrightarrow f^{(m)}, g^{(m)}$ are continuous on $\mathbb{R}$

$\Rightarrow f^{(m)}+g^{(m)}$ is continuous on $\mathbb{R}$

$\therefore \vec{f}+\vec{g}\in C^m(-\infty,\infty)$

Let $k\in\mathbb{R}$, $\vec{f}\in C^m(-\infty,\infty) \Rightarrow f^{(m)}$ is continuous on $\mathbb{R}$

$\therefore k f^{(m)}$ is continuous on $\mathbb{R}$

$\therefore k\vec{f}\in C^m(-\infty,\infty)$

$\therefore C^m(-\infty,\infty)$ is a subspace of $F(-\infty,\infty)$

Notes:

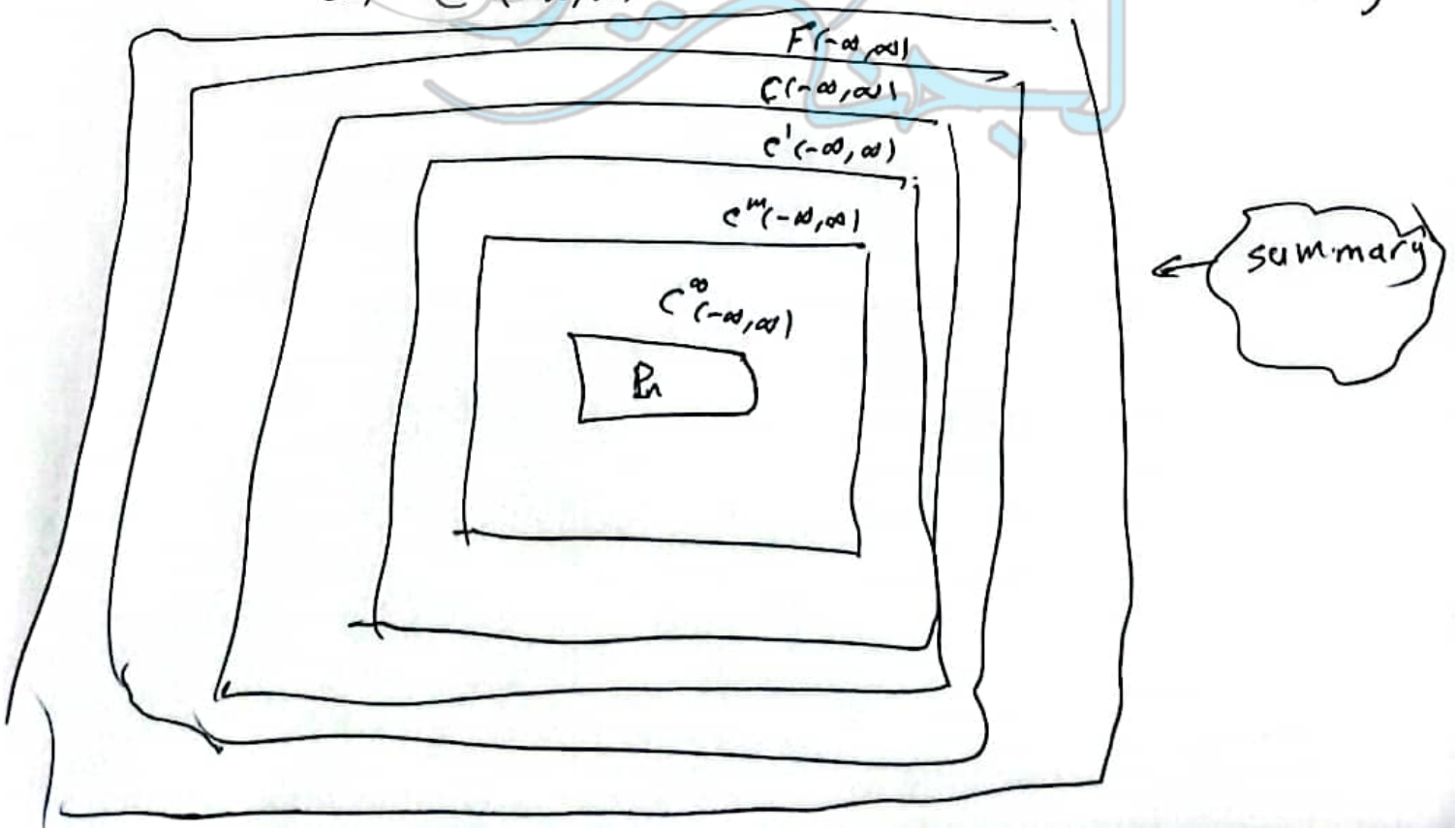
* P_n is a subspace of $C(-\infty,\infty)$
* $C(-\infty,\infty)$ is a subspace of $C^1(-\infty,\infty)$
* $C^1(-\infty,\infty)$ " " " of $C^m(-\infty,\infty)$
* $C^m(-\infty,\infty)$ is a subspace of $F(-\infty,\infty)$

Ex if $C^\infty(-\infty,\infty)$ the of all fuctions that have continuous derivatives of all orders

show that

1) $C^\infty(-\infty,\infty)$ is a subspace of $F(-\infty,\infty)$

2) $C^m(-\infty,\infty)$ is a subspace of $C^\infty(-\infty,\infty)$



e.g: If W_1 and W_2 are two vector subspaces of the vector space V. then

* $W_1 \cap W_2$ is a subspace of V.

Proof:

Let $\vec{u}, \vec{v} \in W_1 \cap W_2$

$\rightarrow \vec{u}, \vec{v} \in W_1$ and $\vec{u}, \vec{v} \in W_2$

$\Rightarrow \vec{u} + \vec{v} \in W_1$ and $\vec{u} + \vec{v} \in W_2$ (why?)

Let $\vec{u} \in W_1 \cap W_2$, $k \in \mathbb{R}$ ... (1)

$\Rightarrow \vec{u} \in W_1$ and $\vec{u} \in W_2$

$\Rightarrow k\vec{u} \in W_1$ and $k\vec{u} \in W_2$ (why?)

$\Rightarrow k\vec{u} \in W_1 \cap W_2$ ... (2)

$\therefore W_1 \cap W_2$ is a subspace of V

Ex: If $W_1 = \left\{ \begin{bmatrix} a_1 & a_2 \\ a_3 & 0 \end{bmatrix} \mid a_1, a_2, a_3 \in \mathbb{R} \right\}$

$W_2 = \left\{ \begin{bmatrix} a_1 & a_2 \\ 0 & a_3 \end{bmatrix} \mid a_1, a_2, a_3 \in \mathbb{R} \right\}$

Show that W_1 is a subspace of M_{22} and W_2 is a subspace of M_{22}

is $W_1 \cup W_2$ a subspace of M_{22} (under the standard operations)

END

Linear Combinations

e.g: If $A\vec{x} = \vec{0}$ is a homogeneous linear system with m-equations and n-unknowns (i.e A is mxn-matrix) then the set of all solutions is a subspace of $\mathbb{R}^n$ under the usual operations.

Proof:

Let $\vec{x}, \vec{y}$ be any two solutions of the system $A\vec{x} = \vec{0}$

$\Rightarrow A\vec{x} = \vec{0}$ and $A\vec{y} = \vec{0}$

$\Rightarrow A\vec{x} + A\vec{y} = \vec{0}$

$\Rightarrow A(\vec{x} + \vec{y}) = \vec{0}$

$\therefore \vec{x} + \vec{y}$ is a solution of $A\vec{x} = \vec{0}$

Now Let $\vec{x}$ be a solution for the system $A\vec{x} = \vec{0}$ and k be a scalar

$\Rightarrow k(A\vec{x}) = k\vec{0}$

$\Rightarrow A(k\vec{x}) = \vec{0}$

$\therefore k\vec{x}$ is a solution for the system $A\vec{x} = \vec{0}$

$\therefore$ the set of all solutions of $A\vec{x} = \vec{0}$ is a subspace of $\mathbb{R}^n$.

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Def:

A vector $\vec{w}$ is called a linear combination of the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r$ if

$\vec{w} = k_1\vec{v}_1 + k_2\vec{v}_2 + \dots + k_n\vec{v}_n$ for some

$k_1, k_2, \dots, k_r \in \mathbb{R}$.

e.g: Let $\vec{u} = (1, 2, -1)$, $\vec{v} = (2, -1, 1)$

$\vec{w} = (5, 0, 1)$, is $\vec{w}$ a linear combination of $\vec{u}$ and $\vec{v}$?

$\vec{w} = k_1\vec{u} + k_2\vec{v}$

$(5, 0, 1) = k_1(1, 2, -1) + k_2(2, -1, 1)$

$(5, 0, 1) = (k_1 + 2k_2, 2k_1 - k_2, -k_1 + k_2)$

$k_1 + 2k_2 = 5$

$2k_1 - k_2 = 0$

$-k_1 + k_2 = 1$

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \\ -1 & 1 & 1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -10 \\ 0 & 3 & 6 \end{array}\right] \begin{array}{l} \\ -2R_1 + R_2 \\ R_1 + R_3 \end{array}$$

$$\sim \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 3 & 6 \end{array}\right] \begin{array}{l} \\ -\frac{1}{2}R_2 \\ \\ \end{array} \sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right] \begin{array}{l} -2R_2 + R_1 \\ \\ -3R_2 + R_3 \end{array}$$

$\Rightarrow k_1 = 1, \; k_2 = 2$

$(5, 0, 1) = 1(1, 2, -1) + 2(2, -1, 1)$

$\Rightarrow \vec{w} = 1\vec{u} + 2\vec{v} \Rightarrow \vec{w}$ is a linear combination of $\vec{u}$ & $\vec{v}$

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e.g: If $\vec{u} = (a,b,c) \in \mathbb{R}^3$, show that $\vec{u}$ is a linear combination of $\hat{i} = (1,0,0)$, $\hat{j} = (0,1,0)$ and $\hat{k} = (0,0,1)$

Sol:

$\vec{u} = (a,b,c)$

$\Rightarrow (a,b,c) = (a,0,0) + (0,b,0) + (0,0,c)$

$(a,b,c) = a(1,0,0) + b(0,1,0) + c(0,0,1)$

$\therefore \vec{u} = a\hat{i} + b\hat{j} + c\hat{k}$

$\therefore \vec{u}$ is a linear combination of $\hat{i}, \hat{j}$ and $\hat{k}$

i.e every vector in $\mathbb{R}^3$ can be expressed as a linear combination of $\hat{i}, \hat{j}$ and $\hat{k}$.

CamScanner

e.g Express $\vec{w} = (5, 8)$ as a linear combination of $\vec{v_1} = (1, 2)$ and $\vec{v_2} = (3, 4)$

Sol:

Let $\vec{w} = k_1\vec{v_1} + k_2\vec{v_2}$

$$k_1 + 3k_2 = 5$$
$$2k_1 + 4k_2 = 8$$

$$\left[\begin{array}{cc|c} 1 & 3 & 5 \\ 2 & 4 & 8 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & -2 & -2 \end{array}\right] -2R_1 + R_2$$

$$\sim \left[\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & 1 & 1 \end{array}\right] -\frac{1}{2}R_2$$

$$\sim \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right] -3R_2 + R_1$$

$k_1 = 2$, $k_2 = 1$

$\therefore (5, 8) = 2(1, 2) + 1(3, 4)$

i.e $\vec{w} = 2\vec{v_1} + 1\vec{v_2}$

eg : Is $\vec{w} = (0,4,5)$ a linear combination of $\vec{u} = (0,-2,2)$ and $\vec{v} = (1,3,-1)$?

Let $\vec{w} = k_1\vec{u} + k_2\vec{v}$

$$(0,4,5) = k_1(0,-2,2) + k_2(1,3,-1)$$

$$\Rightarrow \quad k_2 = 0$$

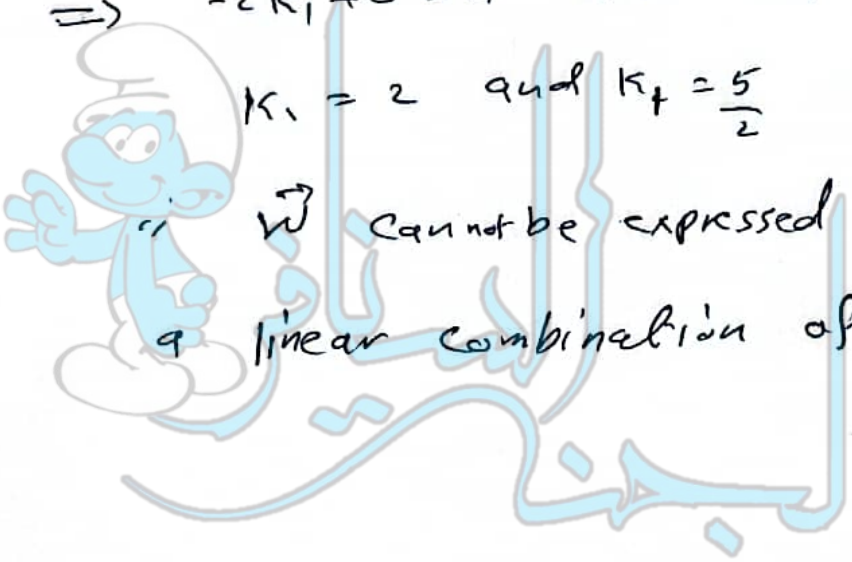
$$-2k_1 + 3k_2 = 4$$

$$2k_1 - k_2 = 5$$

$$\Rightarrow \quad -2k_1 + 0 = 4 \quad \text{and} \quad 2k_1 = 5$$

$k_1 = 2$ and $k_1 = \frac{5}{2}$ contradiction

$\therefore \vec{w}$ cannot be expressed as a linear combination of $\vec{u}$ & $\vec{v}$



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e.g: Can you express $\vec{u} = (1,2)$ as a linear Combination of $\vec{v}_1 = (-1,1)$, $\vec{v}_2 = (2,3)$

$\longrightarrow \vec{u} = K_1\vec{v}_1 + K_2\vec{v}_2$

$(1,2) = K_1(-1,1) + K_2(2,3)$

$\longrightarrow$ $-K_1 + 2K_2 = 1$

$K_1 + 3K_2 = 2$ +

$0 + 5K_2 = 3 \longrightarrow \boxed{K_2 = \frac{3}{5}}$

$K_1 + \frac{9}{5} = 2 \longrightarrow \boxed{K_1 = \frac{1}{5}}$

Thm:

If V is a vector space, $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r \in V$

then $W = span\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r\}$

$= \{\vec{u} \mid \vec{u} = K_1\vec{v}_1 + K_2\vec{v}_2 + \dots + K_r\vec{v}_r\}$

for some $K_1, K_2, \dots, K_r \in \mathbb{R}$

is a vector subspace of V

Dr. Ayham Alfeifel

Omar Waleed
Ayham Asfoor

Proof:

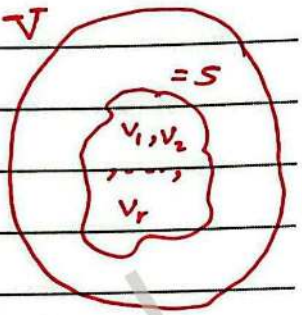
Let $\vec{u}_1, \vec{u}_2 \in W = \text{span}(S)$

$\Rightarrow \vec{u}_1 = K_1\vec{v}_1 + K_2\vec{v}_2 + \cdots + K_r\vec{v}_r$ for some $K_1, K_2, \ldots, K_r \in \mathbb{R}$

$\Rightarrow \vec{u}_2 \in W = \text{span}(S)$

$\exists\, m_1, m_2, \ldots, m_r \in \mathbb{R}$ such that

$\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2 + \cdots + m_r\vec{v}_r$



$\Rightarrow \vec{u}_1 + \vec{u}_2 = (K_1\vec{v}_1 + K_2\vec{v}_2 + \cdots + K_r\vec{v}_r) + (m_1\vec{v}_1 + m_2\vec{v}_2 + \cdots + m_r\vec{v}_r)$

$= (K_1+m_1)\vec{v}_1 + (K_2+m_2)\vec{v}_2 + \cdots + (K_r+m_r)\vec{v}_r$

$\Rightarrow \vec{u}_1 + \vec{u}_2 \in V$

Let $\vec{u} \in \text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_r\}$

$\Rightarrow \vec{u} = K_1\vec{v}_1 + K_2\vec{v}_2 + \cdots + K_r\vec{v}_r$ for some $K_1, K_2, \ldots, K_r \in \text{span}(S)$

Let $m \in \mathbb{R}$

$m\vec{u} = m(K_1\vec{v}_1 + K_2\vec{v}_2 + \cdots + K_r\vec{v}_r)$

$= (mK_1)\vec{v}_1 + (mK_2)\vec{v}_2 + \cdots + (mK_r)\vec{v}_r$

$\Rightarrow m\vec{u} \in \text{span}(S)$

$\therefore W = \text{span}(S)$ is a vector subspace of V

Omar Waleed
Ayham Asfoor

e.g: show that $V = \mathbb{R}^2 = \text{span}\{(1,0), (0,1)\}$ ($\hat{i}$, $\hat{j}$)

→ Let $(a,b) \in \mathbb{R}^2$ be any vector

$(a,b) = a(1,0) + b(0,1)$

$\therefore \text{span}\{\hat{i}, \hat{j}\} = \mathbb{R}^2$

e.g: Prove or disprove

1] $\mathbb{R}^2 = \text{span}\{(1,2), (0,4)\}$

2] $\mathbb{R}^2 = \text{span}\{(1,2), (-2,-4)\}$

1] Let $(a,b) \in \mathbb{R}^2$

$(a,b) = K_1(1,2) + K_2(0,4)$

→ $K_1 = a \in \mathbb{R}$

→ $2K_1 + 4K_2 = b$

$2a + 4K_2 = b$

$K_2 = \frac{b-2a}{4} \in \mathbb{R}$

$\therefore (a,b) = a(1,2) + \frac{b-2a}{4}(0,4)$

$\therefore \mathbb{R}^2 = \text{span}\{(1,2), (0,4)\}$

Omar Waleed
Ayham Asfoor

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2 $\mathbb{R}^2 = \text{span}\{(1,2), (-2,-4)\}$

Let $(a,b) \in \mathbb{R}^2$

$K_1(1,2) + K_2(-2,-4) = (a,b)$

$\rightarrow$ $K_1 - 2K_2 = a$ | (-2)

$2K_1 - 4K_2 = b$

$$\left[\begin{array}{cc|c} 1 & -2 & a \\ 2 & -4 & b \end{array}\right]$$

$$\sim \left[\begin{array}{cc|c} 1 & -2 & a \\ 0 & 0 & -2a+b \end{array}\right] \quad -2R_1 + R_2$$

$b \neq 2a$

$\rightarrow$ (a,b) can be written as a linear combination of $(1,2)$ and $(-2,-4)$

$\therefore \mathbb{R}^2 \neq \text{span}\{(1,2), (-2,-4)\}$

$\mathbb{R}^2$ is not spanned by $\{(1,2), (-2,-4)\}$

Omar Waleed
Ayham Asfoor

e.g: Does $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

span $M_{22} = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \middle| a, b, c, d \in \mathbb{R} \right\}$

under the usual operation

$\rightarrow$ Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{22}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$\therefore S$ spans M_{22}

$$\boxed{M_{22} = \text{span}(S)}$$

Dr. Bahnel Meifel

Omar Waleed
Ayham Asfoor

Linear independence

Def: If V is a vector space

$S = \{\vec{v}_1, \vec{v}_2, \cdots, \vec{v}_r\} \subseteq V$, then

S is called linearly independent if

$$k_1\vec{v}_1 + k_2\vec{v}_2 + \cdots + k_r\vec{v}_r = \vec{0}$$

only when $k_1 = k_2 = k_3 = \cdots = k_r = 0$

but there are constants $k_1, k_2, \cdots, k_r$ not all zeros such

$$k_1\vec{v}_1 + k_2\vec{v}_2 + \cdots + k_r\vec{v}_r = \vec{0}$$

then S is called linearly dependent

e.g: which of the following is a linear independence set in $\mathbb{R}^3$

1. $S = \{(1,2,-1), (0,0,0), (-1,5,6)\}$

$\rightarrow 0(1,2,-1) + 5(0,0,0) + 0(-1,5,6) = (0,0,0)$

$\therefore S$ is linearly dependent

Dr. Ayham Alsafadi

Omar Waleed
Ayham Asfoor

[2] $S = \{(1,2,3), (2,4,6), (-1,1,5)\}$

$\rightarrow -2(1,2,3) + 1(2,4,6) + 0(-1,1,5) = (0,0,0)$

$\therefore$ S is linearly dep.

[3] $S = \{(1,1,1), (0,-1,1), (1,0,-1)\}$

$\rightarrow K_1(1,1,1) + K_2(0,-1,1) + K_3(1,0,-1) = (0,0,0)$

$$K_1 + K_3 = 0$$
$$K_1 - K_2 = 0$$
$$K_1 + K_2 - K_3 = 0$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 1 & -2 & 0 \end{array}\right] \begin{array}{l} \\ -R_1 + R_2 \\ -R_1 + R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -2 & 0 \end{array}\right] -R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right] -R_2 + R_3$$

$-3K_3 = 0 \rightarrow \boxed{K_3 = 0}$

$K_2 + K_3 = 0 \rightarrow \boxed{K_2 = 0}$

$K_1 + K_3 = 0 \rightarrow \boxed{K_1 = 0}$

trivial sol.

$\therefore$ S is linearly indep.

Dr. Ibrahim Al-Hleifel

Omar Waleed
Ayham Asfoor

→ if $\det(A) \neq 0$ → trivial Sol.
→ Linearly indep

if $\det(A) = 0$ → non-trivial sol.
→ Linearly dep

Eg: If $f(-\infty, \infty)$, Let $S = \{2\sin^2 x, \cos^2 x, 1\}$
is S linearly independent?

→ $\overset{K_1}{(1)}(2\sin^2 x) + \overset{K_2}{(2)}(\cos^2 x) + \overset{K_3}{(-2)}(1) = 0$

$2\sin^2 x + 2\cos^2 x - 2 = 0$

∴ S is linearly dep.

Dr. Ahmed Heifel

Omar Waleed
Ayham Asfoor

$S = \{(1,2,-1), (1,0,3)\}$

$2(1,2,-1) + 3(1,0,3)$

$= (2,4,-2) + (3,0,9)$

$= (5,4,7)$

$w = (2,2,5)$

$w = k_1(1,2,-1) + k_2(1,0,3)$

$(2,2,5) = (k_1 + k_2, 2k_1, -k_1 + 3k_2)$

$k_1 + k_2 = 2 \rightleftarrows k_2 = 1$

$2k_1 = 2 \rightarrow k_1 = 1$

$-k_1 + 3k_2 = 5 \rightarrow k_2 = 2$

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 0 & 2 \\ -1 & 3 & 5 \end{array}\right]$$

S is linearly independent

$S = \{(1,1), (2,2), (1,3)\}$

$k_1(1,1) + k_2(2,2) + k_3(1,3) = (0,0)$

$k_1 + 2k_2 + k_3 = 0$

$k_1 + 2k_2 + 3k_3 = 0$

$-2(1,1) + 1(2,2) + 0(1,3) = (0,0)$

Basis of vector space

Def: If S is a subset of the vector space V such that

1) $V = \text{span}(S)$
2) S is linearly independent then S is called a basis of the vector space V

in addition if $|V| = \#(V) = n$ then the dimension of V is n
$(\dim(V) = n)$ (finite dimensional vector space)
else, V is called infinite dimensional vector space

Note: $\dim(\{\vec{0}\}) = 0$

Eg Which of the following is a basis for the vector space

$\mathbb{R}^2 = V$ under the usual operations

1) $S_1 = \{(1,1), (-1,2)\}$

let (a,b) be any vector in $\mathbb{R}^2 = V$

$(a,b) = k_1(1,1) + k_2(-1,2)$

$$\left[\begin{array}{cc|c} 1 & -1 & a \\ 1 & 2 & b \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -1 & a \\ 0 & 3 & b-a \end{array}\right] -R_1+R_2$$

$3k_2 = b-a \rightarrow k_2 = \frac{b-a}{3}$

$k_1 - k_2 = a$

$k_1 - \frac{b-a}{3} = a$

$k_1 = a + \frac{b-a}{3}$

$k_1 = \frac{2a+b}{3}$

$\therefore span(S) = \mathbb{R}^2$

$S_1 = \{(1,1), (-1,2)\}$

$k_1(1,1) + k_2(-1,2) = (0,0)$

$$\begin{bmatrix} 1 & -1 & | & 0 \\ 1 & 2 & | & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & | & 0 \\ 0 & 3 & | & 0 \end{bmatrix} \; -R+R_2$$

$3k_2 = 0 \rightarrow k_2 = 0$

$k_1 - k_2 = 0 \rightarrow k_1 = k_2 = 0$

$\therefore S_1$ is linearly independent

$\therefore S_1$ is a basis for $\mathbb{R}^2$

فهو لشفاء وإن من مرض

2) $S_2 = \{(1,2), (2,4)\}$

$-2(1,2) + 1(2,4) = (0,0)$

$\therefore S_2$ is linearly dependent

$\therefore S_2$ is not a basis for $\mathbb{R}^2$

3) $S_3 = \{(1,2), (-1,0), (1,3)\}$

$\dim(\mathbb{R}^2) = 2 \neq |S_3|$

$\therefore S_3$ is not a basis for $\mathbb{R}^2$

4) $S_4 = \{(1,1)\}$

$\therefore S_4$ is not a basis $\mathbb{R}^2$ since $\dim(\mathbb{R}^2) = 2 \neq 1 = |S_4|$

Thm: If V is a finite dimensional vector space, $S \subseteq V$ then S is linearly independant if & $V = \text{span}(S)$

Important basis for some vector spaces

1) $V = \mathbb{R}^2$ $(\dim(\mathbb{R}^2) = 2)$

$S = \{(1,0), (0,1)\}$

$= \{\vec{i}, \hat{j}\}$

2) $V = \mathbb{R}^3$ $(\dim(\mathbb{R}^3) = 3)$

$S = \{\hat{i}, \hat{j}, \hat{k}\}$

$= \{(1,0,0), (0,1,0), (0,0,1)\}$

3) $V = \mathbb{R}^n$ $(\dim(\mathbb{R}^n) = n)$

$S = \{e_1, e_2, \ldots, e_n\}$

$= \{(1,0,\ldots,0), (0,1,0,\ldots,0), \ldots, (0,0,\ldots,0,1)\}$

4) $V = M_{nm}$ $\dim(V) = nm$

$S = \{A \mid A \text{ is } n \times m \text{ matrix}\}$

where $A = (a_{st}) = \begin{cases} 0, & \forall s \neq i, \forall t \neq j \\ 1, & s = i, t = j \end{cases}$

5) $V = M_{22} = \left\{A \mid A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, a, b, c, d \in \mathbb{R}\right\}$

$\dim(V) = 4$

$S = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\}$

6) $P_n = \{f \mid f = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n,\ a_0, a_1, \ldots, a_n \in \mathbb{R}\}$

$\dim(P_n) = n+1$

$S = \{1, x, x^2, \ldots, x^n\}$

$P_2 = \{f \mid f = a_0 + a_1 x + a_2 x^2\}$

$S = \{1, x, x^2\}$

$2x^2 - 2 = 0(1) + (-1)\pi + 2x^2$

Eg. Which of the following is a basis for P_2

1) $S_1 = \{t, t^2+1\}$

S_1 is not a basis for P_2

2) $S_2 = \{1, t, t^2+1\}$

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\det(A) = 1 \neq 0$

$\therefore S_1$ is linearly independent

$\therefore S_1$ is a basis for P_2

minus/plus theorem

If S is a non-empty subset of the vector space V then:

1) if S is linearly independent and $\vec{v} \notin \text{span}(S)$ then $S \cup \{\vec{v}\}$ is linearly independent

2) if $\vec{v} \in \text{span}(S)$ then $\text{span}(S) = \text{span}(S \setminus \{\vec{v}\})$

Eigenvalues and Eigenvectors

Def: If A is $n \times n$-matrix
x is a non-zero vector
in R^n then
x is called an eigen vector of A
if λ is a scalar such that
$Ax = \lambda x$
λ is called an eigenvalue & x
is corresponding eigenvector

$$Ax = \lambda x$$
$$\lambda x - Ax = \vec{0}$$
$$(\lambda I - A)x = \vec{0}$$

$$|\lambda I - A| = 0 \qquad (\det(\lambda I - A) = 0)$$

↳ eigenvalues

$P(\lambda)$ = characteristic polynomial

Eg. Find the eigenvalues and eigenvectors of

1) $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

$$\lambda I - A = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} \lambda - 1 & -4 \\ -2 & \lambda - 3 \end{bmatrix}$$

$$|\lambda I - A| = (\lambda-1)(\lambda-3) - 8$$
$$= \lambda^2 - 4\lambda + 3 - 8$$
$$= \lambda^2 - 4\lambda - 5$$

$P(\lambda) = \lambda^2 - 4\lambda - 5$ ⟶ Characteristic Polynomial

$\lambda^2 - 4\lambda - 5 = 0$

$(\lambda - 5)(\lambda + 1) = 0$

$\lambda_1 = 5, \lambda_2 = -1$ eigenvalues

$\lambda_1 = 5, \; x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ $\quad (\lambda I - A)\vec{x} = 0$

$$\begin{bmatrix} 4 & -4 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$4x_1 - 4x_2 = 0 \longrightarrow x_1 = x_2$

$-2x_1 + 2x_2 = 0$

$\therefore x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is the corresponding eigenvector

$\lambda_2 = -1, \; y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$$\begin{bmatrix} -2 & -4 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-2y_1 - 4y_2 = 0 \longrightarrow y_1 = -2y_2$

$-2y_1 - 4y_2 = 0$

$y = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ eigenvector

2) $A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$

$|\lambda I - A| = 0$

$\begin{bmatrix} \lambda-2 & -3 \\ 0 & \lambda-2 \end{bmatrix} = 0$

$(\lambda-2)(\lambda-2) - 0 = 0$

$(\lambda-2)^2 = 0$

$\rightarrow \lambda_1 = 2, \lambda_2 = 2$

$\lambda_1 = \lambda_2 = 2$, Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\begin{bmatrix} 0 & -3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-3x_2 = 0$

$x_2 = 0$

$\therefore x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ eigen vector

3) $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

$|\lambda I - A| = 0$

$\begin{vmatrix} \lambda-4 & 0 & -1 \\ 2 & \lambda-1 & 0 \\ 2 & 0 & \lambda-1 \end{vmatrix} = 0$

$$(\lambda-4)\left((\lambda-1)^2-0\right)-0+(-1)\left(0-2(\lambda-1)\right)=0$$

$$(\lambda-1)\left((\lambda-4)(\lambda-1)+2\right)=0$$

$$(\lambda-1)(\lambda^2-5\lambda+4+2)=0$$

$$(\lambda-1)(\lambda^2-5\lambda+6)=0$$

$$(\lambda-1)(\lambda-2)(\lambda-3)=0$$

$$\lambda_1=1,\ \lambda_2=2,\ \lambda_3=3$$

$$\lambda_1=1,\quad x=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$$

$$\begin{bmatrix}-3&0&-1\\2&0&0\\2&0&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\left[\begin{array}{ccc|c}-3&0&-1&0\\2&0&0&0\\2&0&0&0\end{array}\right]\sim\left[\begin{array}{ccc|c}1&0&0&0\\-3&0&-1&0\\2&0&0&0\end{array}\right]\begin{matrix}\frac{1}{2}R_2\\R_1\end{matrix}$$

$$\left[\begin{array}{ccc|c}1&0&0&0\\0&0&-1&0\\0&0&0&0\end{array}\right]\begin{matrix}3R_1+R_2\\-2R_1+R_3\end{matrix}\sim\left[\begin{array}{ccc|c}1&0&0&0\\0&0&1&0\\0&0&0&0\end{array}\right]-R_2$$

$$x_1=0$$
$$x_2=0$$
$$x_3=1$$

$$\therefore x=\begin{bmatrix}0\\1\\0\end{bmatrix}$$

If $A\vec{x} = \lambda\vec{x}$

($\vec{x}$: eigenvector, λ: eigenvalue)

$$A\vec{x} = \lambda\vec{x}$$
$$\lambda(A\vec{x}) = \lambda^2\vec{x}$$
$$(A\lambda\vec{x}) = \lambda^2\vec{x}$$
$$(AA\vec{x}) = \lambda^2\vec{x}$$
$$A^2\vec{x} = \lambda^2\vec{x}$$

λ^2 is an eigenvalue of A^2

$$A^2\vec{x} = \lambda^2\vec{x}$$
$$*\lambda \rightarrow \lambda(A^2\vec{x}) = \lambda^3\vec{x}$$
$$(A^2\lambda\vec{x}) = \lambda^3\vec{x}$$
$$A^2A\vec{x} = \lambda^3\vec{x}$$
$$A^3\vec{x} = \lambda^3\vec{x}$$

λ^3 is an eigenvalue of A^3

λ^n is an eigenvalue of A^n

$\vec{x}$: eigenvector

فَخَلَفَ مِنْ بَعْدِهِمْ خَلْفٌ أَضَاعُوا الصَّلَوٰةَ
وَاتَّبَعُوا الشَّهَوَاتِ ۖ فَسَوْفَ يَلْقَوْنَ غَيًّا

Linear Transformation

Def: If V and W are two vector spaces then the function

$$T: V \to W$$

is called a linear transformation if:

1) $\forall \vec{u}, \vec{v} \in V$

$$T(\vec{u}+\vec{v}) = T(\vec{u}) + T(\vec{v})$$

2) $\forall \vec{u} \in V \quad \forall k \in R$

$$T(k\vec{u}) = k\,T(\vec{u})$$

Eg. Which of the following is a linear transformation (L.T)

1) $T: R^2 \to R^2$

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ x-y \end{bmatrix}$$

Let $\vec{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \in R^2$

$$T(\vec{u}+\vec{v}) = T\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} x_1+x_2 \\ y_1+y_2 \end{bmatrix}\right)$$

$$= \begin{bmatrix} (x_1+x_2)+(y_1+y_2) \\ (x_1+x_2)-(y_1+y_2) \end{bmatrix} = \begin{bmatrix} (x_1+y_1)+(x_2+y_2) \\ (x_1-y_1)+(x_2-y_2) \end{bmatrix}$$

$$= \begin{bmatrix} x_1 + y_1 \\ x_1 - y_1 \end{bmatrix} + \begin{bmatrix} x_2 + y_2 \\ x_2 - y_2 \end{bmatrix}$$

$$= T\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + T\begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$= T(\vec{u}) + T(\vec{v})$$

let $\vec{u} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$

$k \in \mathbb{R}$

$$T(k\vec{u}) = T\left(k\begin{bmatrix} x \\ y \end{bmatrix}\right)$$

$$= T\begin{bmatrix} kx \\ ky \end{bmatrix}$$

$$= \begin{bmatrix} kx + ky \\ kx - ky \end{bmatrix}$$

$$= k\begin{bmatrix} x + y \\ x - y \end{bmatrix}$$

$$= k\,T(\vec{u})$$

$\therefore T$ is a L.T

Note: $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ x-y \end{bmatrix}$

$A: \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$T\begin{bmatrix} x \\ y \end{bmatrix} = A\begin{bmatrix} x \\ y \end{bmatrix}$

↑ matrix of transformations

2) $T: P_1 \to P_2$

$f = ax+b$

$T(f) = ax^2+bx$

Let $f = a_1x+b_1$, $g = a_2x+b_2$

$T(f+g) = T[(a_1+a_2)x+(b_1+b_2)]$

$= (a_1+a_2)x^2+(b_1+b_2)x$

$= a_1x^2+b_1x+a_2x^2+b_2x$

$= T(f)+T(g)$

Let $f = ax+b \in P_1$ $\quad k \in \mathbb{R}$

$T(kf) = T(k(ax+b))$

$= T(kax+kb)$

$= kax^2+kbx$

$= k(ax^2+bx)$

$= kT(f)$ $\quad \therefore T$ is a L.T

3) $T: R^2 \to R^2$

$T(x,y) = (x+y, y+1)$

let $(x_1,y_1), (x_2,y_2) \in R^2$

$T((x_1,y_1)+(x_2,y_2)) = T((x_1+x_2, y_1+y_2))$

$= (x_1+x_2+y_1+y_2, y_1+y_2+1)$

$T(x_1,y_1) + T(x_2,y_2) = (x_1+y_1, y_1+1) + (x_2+y_2, y_2+1)$

$= (x_1+x_2+y_1+y_2, y_1+y_2+2)$

$\therefore T(\vec{u}+\vec{v}) \neq T(\vec{u}) + T(\vec{v})$

$\therefore T$ is not a L.T

4) $T: M_{22} \to \mathbb{R}$

$T(A) = \det(A)$

$T(A+B) = \det(A+B)$

$T(A) + T(B) = \det(A) + \det(B)$

$T(A+B) \neq T(A) + T(B)$ in general

$\therefore T$ is not a linear transformation

5) $T: C^1(-\infty, \infty) \to \mathbb{R}$

$T(f) = \int_a^b f(t)\,dt$

let $f, g \in C^1(-\infty, \infty)$

$T(f+g) = \int_a^b (f(t)+g(t))\,dt = \int_a^b f(t)\,dt + \int_a^b g(t)\,dt$

$= T(f) + T(g)$

$\therefore T$ is a L.T

let $k \in \mathbb{R}$, $f \in C^1(-\infty, \infty)$

$$T(kf) = \int_a^b (kf)(t)\,dt$$

$$= \int_a^b k f(t)\,dt$$

$$= k \int_a^b f(t)\,dt$$

$$= k\,T(f)$$

$\therefore T$ is a L.T

Thm: If V and W are two vector spaces then $T: V \to W$

1) $T(\vec{0}_V) = \vec{0}_W$
2) $T(-\vec{u}) = -T(\vec{u}) \quad \forall \vec{u} \in V$
3) $T(\vec{u} - \vec{v}) = T(\vec{u}) - T(\vec{v}) \quad \forall \vec{u}, \vec{v} \in V$

Proof:

1) $(0)\,\vec{0}_V = \vec{0}_{\vec{u}}$

$T(\vec{0}_V) = T(0\,\vec{0}_V)$

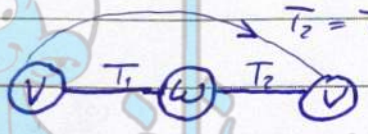
$= 0\,T(\vec{0}_V)$

$= \vec{0}$

2) $T(-\vec{u}) = T((-1)\vec{u})$
$= (-1)\,T(\vec{u})$
$= -T(\vec{u})$

3) $T(\vec{u}-\vec{v}) = T(\vec{u}+(-\vec{v}))$
$= T(\vec{u}) + T(-\vec{v})$
$= T(\vec{u}) - T(\vec{v})$

Thm: If $T_1: V \to W$ is a L.T
and $T_2: W \to U$ is a L.T
then $T_2 \circ T_1: V \to U$ is a L.T



• Proof:
let $\vec{u}, \vec{v} \in V$
$T_2 \circ T_1(\vec{u}+\vec{v}) = T_2(T_1(\vec{u}+\vec{v}))$
$= T_2(T_1(\vec{u}) + T_1(\vec{v}))$ T_1 is a L.T
$= T_2(T_1(\vec{u})) + T_2(T_1(\vec{v}))$ T_2 is a L.T
$= (T_2 \circ T_1)(\vec{u}) + (T_2 \circ T_1)(\vec{v})$

let $k \in \mathbb{R}$ $\vec{u} \in V$
$(T_2 = T_1)(k\vec{u}) = T_2(T_1(k\vec{u}))$
$= T_2(k\,T_1(\vec{u}))$ T_1 is L.T
$= k\,T_2(T_1(\vec{u}))$ T_2 is L.T
$= k(T_2 + T_1)(u)$
$\therefore T_2 + T_1$ is
a linear transformation

$f: \mathbb{R} \to \mathbb{R}$

$f(x) = x^2$

range $= [0, \infty)$ f is not onto f is not 1-1

$f: \mathbb{R} \to [0, \infty)$

$f(x) = x^2$

range $= [0, \infty)$ = co-domain

f is onto

f is not 1-1

$f: R \to \mathbb{R}$

$f(x) = 2x$

range $(f) = \mathbb{R}$ = co-domain

onto

f is 1-1

f is 1-1 and bijective

1-1 = injective

onto = surjective

1-1 + onto = bijective

Onto and 1-1 Transformations

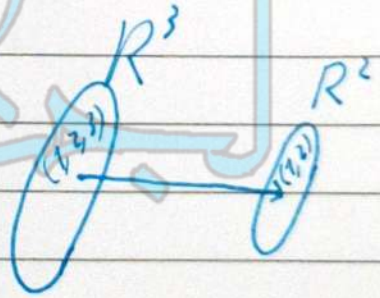
Def. If V and W are two vector spaces,
$T: V \to W$ is a L.T

then

1) T is called 1-1 if $T(\vec{v}) = T(\vec{w}) \to \vec{v} = \vec{w}$
for any $\vec{v}, \vec{w} \in V$
1-1 ≡ injective

2) T is onto if $\forall \vec{w} \in W, \exists \vec{u} \in V$
such that $f(\vec{u}) = \vec{v}$
or (range $\to f(v) = w$)
onto ≡ surjective

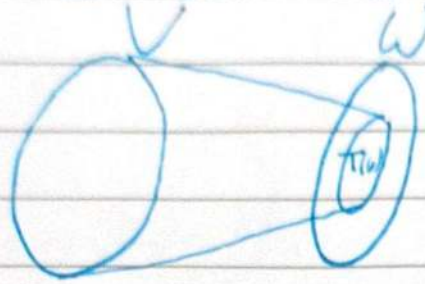
E.g. If $T: \mathbb{R}^3 \to \mathbb{R}^2$
$T(x, y, z) = (x, y)$
is T 1-1?
is T onto?



★ $T(1,1,2) = (1,1)$
$T(1,1,3) = (1,1)$
$(1,1,2) \neq (1,1,3)$ but $T(1,1,2) = T(1,1,3)$
∴ T is not 1-1

★ Let $(a,b) \in R^2$
$T(a,b,1) = (a,b) \to \exists (a,b,1) \in R^3$
such that $T(ab,1) = (a,b)$
∴ T is onto

Thm: If $T: V \to W$ is a L.T
then $T(V)$ is a subspace of W



- Proof:

let $\vec{w_1}, \vec{w_2} \in T(V)$

$\to \vec{v_1}, \vec{v_2} \in V$ such that

$T(\vec{v_1}) = \vec{w_1}$ & $T(\vec{v_2}) = \vec{w_2}$

$\vec{w_1} + \vec{w_2} = T(\vec{v_1}) + T(\vec{v_2})$

$= T(\vec{v_1} + \vec{v_2})$

$\to \exists\ \vec{v_1} + \vec{v_2} \in V$ such that

$T(\vec{v_1} + \vec{v_2}) = \vec{w_1} + \vec{w_2}$

$\therefore \vec{w_1} + \vec{w_2} \in T(V)$

let $\vec{w} \in T(V) \to \exists \vec{v} \in V$

such that $T(\vec{v}) = \vec{w}$

let $k \in \mathbb{R}$

$k\vec{w} = k\,T(\vec{r})$

$k\vec{w} = T(k\vec{r})$

$\hookrightarrow \in V$

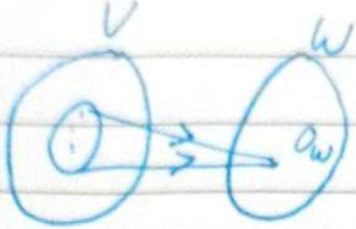
$\therefore k\vec{w} \in T(V)$

$\therefore T(V)$ is a subspace of W

Def: If $T: V \to W$ is a L.T then

$kernel(T) = ker(T)$

$= \{\vec{v} \in V \mid T(\vec{v}) = 0_W\}$



Thm: If $T: V \to W$ is L.T then $ker(T)$ is a subspace of V

- **Proof:**

let $\vec{v_1}, \vec{v_2} \in ker(T)$

$\to T(\vec{v_1}) = 0,\ T(\vec{v_2}) = 0$

$T(\vec{v_1}) + T(\vec{v_2}) = 0 + 0 = 0$

$T(\vec{v_1} + \vec{v_2}) = 0$

$\therefore \vec{v_1} + \vec{v_2} \in ker(T)$

let $k \in \mathbb{R}$, $\vec{v} \in ker(T)$

$\to T(\vec{v}) = 0$

$k\,T(\vec{v}) = (k)(0)$

$T(k\vec{v}) = 0$

$\to k\vec{v} \in ker(T)$

$ker(T)$ is a subspace of V

Thm: If $T: V \to W$ is a L.T
then T is 1-1 if f $\ker(T) = \{0\}$
$\longleftrightarrow$

- Proof:

($\Rightarrow$) T is 1-1
let $\vec{u} \in \ker(T) \to T(\vec{u}) = \vec{0}$
also $T(\vec{0}) = \vec{0}$
$\to T(\vec{u}) = T(\vec{0})$
$\to \vec{u} = \vec{0}$
$\therefore \ker(T) = \{0\}$

($\Leftarrow$) let $\ker(T) = \{\vec{0}\}$
let $\vec{u}, \vec{v} \in V$ such that
$T(\vec{u}) = T(\vec{v})$
$\to T(\vec{u}) - T(\vec{v}) = 0$
$\to T(\vec{u} - \vec{v}) = 0$
$\to \vec{u} - \vec{v} \in \ker(T)$
$\to \vec{u} - \vec{v} = \vec{0}$
$\to \vec{u} = \vec{v}$
$\therefore T$ is 1-1

Eg. If $J: P_1 \to \mathbb{R}$
$J(f) = \int_{-1}^{1} f(x)\,dx$ (L.T)

is J 1-1
$J(0) = 0$
If $f = x$ $J(x) = \int_{-1}^{1} x\,dx = 0$
$\ker(T) \neq \{0\}$
$\therefore T$ is not 1-1

Eg. If A is 2×2-matrix prove that the characteristic function $P(\lambda) = \lambda^2 - \lambda\, tra(A) + det(A)$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$P(\lambda) = \begin{vmatrix} \lambda - a & -b \\ -c & \lambda - d \end{vmatrix}$$

$$= (\lambda - a)(\lambda - d) - bc$$

$$= \lambda^2 - a\lambda - d\lambda + ad - bc$$

$$= \lambda^2 - \lambda(a + d) + det(A)$$

$$= \lambda^2 - \lambda\, tr(A) + det(A)$$